

Kernel-based potential mean-field games with unbiased random Fourier U -statistics

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Abstract

We study the subclass of potential mean-field games in which the running interaction cost and the terminal target cost are both expressed through reproducing-kernel maximum mean discrepancy (MMD) penalties, and develop a computational framework that exploits this kernel structure. Both costs are estimated from finite-sample empirical distributions using a random Fourier U-statistic representation that is unbiased and has linear cost in the batch size. The drift of the controlled diffusion is parametrized by a neural network and trained via stochastic gradient descent. For this subclass we prove a sample-level almost-sure convergence theorem and an explicit almost-sure rate of convergence, under coupled rate conditions on the penalty parameter, the random-feature count, the sample size, and the optimization tolerance. The framework includes the kernel-MMD-penalty Schrödinger bridge problem as the special case of a vanishing interaction cost. Numerical experiments illustrate the method on the Schrödinger bridge problem in dimensions up to one hundred, and on an electric vehicle charging coordination problem with per-vehicle physical heterogeneity, where an aggregate-demand congestion cost represents price-feedback competition at the population level and the terminal MMD penalty shapes the state-of-charge distribution at the deadline.

Key words: mean-field games, maximum mean discrepancy, random Fourier features, U-statistics, neural SDE, Schrödinger bridge.

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1 Introduction

Mean-field games (MFGs) [17, 16] model large populations of strategic agents through the interaction of a representative agent with the population distribution; see the monographs [6, 7] for a comprehensive account. The subclass of *potential* (or *cooperative*) MFGs, where the running and terminal costs admit variational potentials with respect to the population law, reduces to a

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single mean-field stochastic optimal control problem and has been used to model swarm coordination, traffic flow, opinion dynamics, and energy systems (see, e.g., [11, 21, 26]). A particularly important variant is the *planning* (or *coordination*) MFG of Lasry–Lions and Achdou–Camilli–Capuzzo–Dolcetta [1], in which the terminal cost is a distributional functional of the population law at the deadline, typically enforcing a prescribed terminal distribution μ_T as a hard constraint $\mathcal{L}(X_T) = \mu_T$. In its potential / variational form, this problem is equivalently known as *mean-field optimal transport* (MFOT) [3, 5]. A quadratic kinetic energy with an additional running mean-field cost is minimised over flows that transport μ_0 to μ_T , and the Schrödinger bridge of [24] arises as its entropic specialisation with vanishing interaction. Our framework provides a sample-level numerical scheme for the *soft-target* version of MFOT, where the hard planning constraint is relaxed by the kernel-MMD penalty $\lambda\gamma_K^2(\mathcal{L}(X_T), \mu_T)$ introduced below.

A motivating example from the smart-grid literature is coordinated charging of electric vehicle (EV) fleets [11]. A large number of EVs decide, over an operational horizon $[0, T]$, when and how fast to charge. Each EV faces a private charging effort and is exposed to a market electricity price that depends on the aggregate demand of the fleet (the population distribution of charging power), producing a congestion-type cost. At the deadline T , the fleet should reach a desired distribution of state-of-charge. For example, all vehicles charged to a sufficient level for the next day. This is a soft target on the terminal population law $\mathcal{L}(X_T)$ (we write $\mathcal{L}(X)$ for the probability distribution of a random variable X throughout this paper), rather than on individual trajectories. In a representative-agent formulation, this leads to the potential MFG

$$(1.1) \quad \inf_{u \in \mathcal{A}} \int_0^T \left(\frac{1}{2} \mathbb{E}[|u(t, X_t)|^2] + \mathcal{R}[\mathcal{L}(X_t)] \right) dt + \mathcal{T}[\mathcal{L}(X_T)],$$

where X is a controlled diffusion driven by the drift u , $\mathcal{R}$ encodes the congestion cost, and $\mathcal{T}$ encodes the deadline target. Classical grid-based numerical methods for MFGs, e.g., HJB–Fokker–Planck coupled solvers [2, 1] and related schemes surveyed in [18], are limited to dimensions $d \leq 5$ by the curse of dimensionality, whereas deep-learning approaches based on neural backward SDEs, fictitious play, or alternating-network architectures [12, 8, 28, 20] have demonstrated higher-dimensional capability empirically but typically lack rigorous convergence guarantees in the sample-based regime where they are actually trained.

In this paper, we develop a computational framework for the potential MFG (1.1) where both the running interaction cost $\mathcal{R}$ and the terminal cost $\mathcal{T}$ are expressed through reproducing-kernel maximum mean discrepancies (MMDs). The per-iteration cost of the empirical objective is linear in the sample size N via the random Fourier U -statistic representation introduced in Section 3, in contrast to the $O(N^2)$ cost of the original kernel-MMD-penalty approach of [24]. Specifically, we take

$$(1.2) \quad \mathcal{R}[\nu_t] = \frac{c}{2} \iint W(x, y) \nu_t(dx) \nu_t(dy), \quad \mathcal{T}[\nu_T] = \lambda\gamma_K(\nu_T, \mu_T)^2,$$

where W is a non-negative symmetric kernel describing the congestion. We consider two instantiations: a translation-invariant kernel (e.g., the Gaussian) when an $O(NM)$ random-feature representation is desired (Section 3.2), and the rank-one aggregate-demand kernel of Section 5.6 for the EV application. The functional γ_K is the MMD associated with a characteristic kernel K

that relaxes the deadline constraint $\nu_T = \mu_T$. The kernel choice is motivated by the kernel-based Schrödinger bridge problem (SBP) studied by the author in [24], which corresponds to the special case $\mathcal{R} \equiv 0$ of (1.1)–(1.2); in that work the empirical kernel U -statistic of (1.3) was used, restricting the experiments to dimensions $d \leq 2$ due to the $O(N^2)$ kernel-evaluation cost.

$$(1.3) \quad \bar{\gamma}_K^2 = \frac{1}{N(N-1)} \sum_i \sum_{j \neq i} K(X_i, X_j) - \frac{2}{N^2} \sum_{i,j} K(X_i, Y_j) + \frac{1}{N(N-1)} \sum_i \sum_{j \neq i} K(Y_i, Y_j).$$

To overcome this scaling barrier, we use random Fourier features (RFF) [27]. Standard RFF-based estimators of the MMD² have been studied for hypothesis testing [37, 33, 10] and take the form of a V -statistic with positive bias of order $O(\Phi(0)/N)$. This bias is asymptotically negligible in testing but introduces a persistent distortion at the population level that prevents a clean limit-theorem analysis as the MMD penalty weight grows. We instead construct an *unbiased* $O(NM)$ estimator of γ_K^2 by combining the Fourier representation

$$\gamma_K(\mu, \nu)^2 = \Phi(0) \int |\tilde{\mu}(\xi) - \tilde{\nu}(\xi)|^2 \rho(\xi) d\xi$$

with a U -statistic decomposition that eliminates the diagonal bias in $O(N)$ per frequency. The same construction applies to the congestion energy $\mathcal{R}[\nu_t]$ when the kernel W is translation-invariant with positive Fourier transform, yielding an unbiased $O(NM)$ estimator of the interaction cost (Theorem 3.3). For congestion energies of aggregate-demand type used in Section 5.6, a simpler unbiased $O(N)$ U -statistic is available without random features (see (5.4)). The drift u is parametrized by a neural network and trained by stochastic gradient descent on the empirical version of (1.1); the empirical interaction cost $\hat{\mathcal{R}}[\hat{\nu}_t^N]$ and the empirical terminal penalty $\lambda \hat{\gamma}_{M,N}^2(\hat{\nu}_T^N, \mu_T)$ are evaluated on the current minibatch at $O(NM)$ cost (or $O(N)$ for the aggregate-demand congestion).

Relation to existing computational MFG methods. The HJB–Fokker–Planck coupled approach of Achdou and collaborators [2, 1] provides rigorous numerical schemes for MFGs but is restricted to dimensions $d \leq 5$ by the curse of dimensionality. Variational primal methods of generalised conditional-gradient (Frank–Wolfe) type for potential MFGs have been developed by Nakamura and Saito [23, 22], with explicit non-asymptotic convergence rates established at the fully discretised level on grids; these analyses cover local-coupling running costs but are tied to finite-difference discretisations. Deep-learning MFG methods have addressed the high-dimensional regime through several routes: the alternating population–control networks of [20] (APAC-Net) and the related framework of [28], fictitious-play and policy-iteration variants surveyed in [8, 9], and bridge-matching or diffusion-Schrödinger-bridge methods [12, 35, 31, 34] that solve the $c = 0$ specialization of (1.1). These methods have demonstrated empirical scalability, but typically rely on training-objective heuristics or auxiliary networks and lack sample-level convergence guarantees in the regime in which they are actually trained. Our method takes a structurally simpler route within the kernel-MMD terminal-penalty subclass: a single end-to-end optimization with one drift network and a closed-form unbiased $O(NM)$ estimator of the terminal MMD penalty, together with a flexible empirical estimator of the running interaction cost (unbiased $O(NM)$ for kernel-MMD interactions).

and unbiased $O(N)$ for aggregate-demand costs), avoiding both the alternating optimization between population and control networks and the auxiliary score networks of bridge-matching schemes. This structural simplicity is what enables the sample-level almost-sure convergence theorem and the explicit almost-sure rate of Section 4. We illustrate the method at dimensions up to $d = 100$ for the special case $\mathcal{R} \equiv 0$ (Schrödinger bridge), and on an electric-vehicle charging coordination problem with a physical per-vehicle heterogeneity, where the soft MMD target shapes the deadline state-of-charge distribution and the aggregate-demand running cost reduces peak and time-averaged grid load.

Scope. Our framework targets the soft-target mean-field optimal transport problem in which the terminal cost is the squared MMD against a prescribed distribution μ_T . This structure makes the soft deadline penalty $\lambda\gamma_K^2(\nu_T, \mu_T)$ available as a relaxation of the planning constraint $\nu_T = \mu_T$, and enables the unbiased $O(NM)$ random-feature representation of γ_K^2 that underlies the convergence guarantees of Section 4. The running mean-field cost $\mathcal{R}[\nu_t]$ is more flexible. The convergence theorem only requires $\mathcal{R}$ to be non-negative, weakly continuous on $\mathcal{P}(\mathbb{R}^d)$, and accompanied by a strongly consistent empirical estimator (Proposition 4.8). This covers both kernel-MMD interactions for which we give an unbiased $O(NM)$ random-feature estimator in Section 3.2 and aggregate-demand costs of the form $D[\nu]^2$ used in the EV application of Section 5.6. Extending to weakly lower-semicontinuous functionals such as local-density costs $\int F(\rho(x))dx$ is a natural next step.

Contributions.

- (i) We derive a Fourier representation of γ_K^2 and of the kernel self-interaction $\iint W(x, y)\nu(dx)\nu(dy)$, and construct unbiased U -statistic estimators $\hat{\gamma}_{M,N}^2$ for γ_K^2 (Theorem 3.1) and $\hat{\mathcal{R}}_{M,N}[\hat{\nu}^N]$ for $\mathcal{R}[\nu]$ (Theorem 3.3), each of complexity $O(NM)$ and admitting an explicit variance decomposition $O(1/M) + O(1/N)$.
- (ii) We formulate the empirical kernel-MMD soft-target MFOT problem (Section 4.1) and establish a sample-level almost-sure convergence theorem for it (Theorem 4.6), with the Schrödinger-bridge specialization recovered as Corollary 4.7. Both the energy and the kernel costs are estimated from finite samples, and the sequences $\lambda_n, M_n, N_n, \varepsilon_n$ are coupled via the explicit rate conditions $\lambda_n \rightarrow \infty$, $M_n/\log n \rightarrow \infty$, $N_n \rightarrow \infty$, $\varepsilon_n \rightarrow 0$, $\lambda_n \varepsilon_n \rightarrow 0$. Under a uniform bound on the drift, we further obtain an explicit almost-sure rate of the form $\leq (\text{constant} + \varepsilon_n)/\lambda_n + C^*(\sqrt{\log n/M_n} + \sqrt{\log n/N_n})$ holding for all sufficiently large n , for the terminal discrepancy $\gamma_K(\mathcal{L}(X_T^{(n)}), \mu_T)^2$, yielding an algebraic rate $\gamma_K = O(n^{-a/2}(\log n)^{1/4})$ under polynomial schedules with $\lambda_n = n^a$ and $M_n = N_n = n^{2a}$ (Proposition 4.9 for the SBP case and Proposition 4.11 for the MFG case).
- (iii) The Schrödinger bridge problem of [24] is recovered as the special case $\mathcal{R} \equiv 0$ (Corollary 4.7), and taking the $M = \infty$ limit yields a sample-level convergence theorem for the kernel-MMD penalty method of [24] (Corollary 4.12). To the author’s knowledge, this is the first sample-level convergence guarantee for the empirical implementation of that method. The convergence theorem itself extends beyond the kernel-MMD structure of the running cost to any non-negative weakly continuous functional with a consistent empirical estimator (Proposition 4.8),

which we use in Section 5.6 to handle an aggregate-demand congestion cost of the form $D[\nu]^2 = (\int u^*(s)\nu(ds))^2$.

- (iv) Numerical experiments cover the Schrödinger bridge specialization in dimensions up to $d = 100$ (Gaussian shift and bimodal targets; Sections 5.2–5.5) and the EV charging fleet potential MFG with a physical per-vehicle heterogeneity (log charging-speed multiplier; Section 5.6), demonstrating that the framework simultaneously recovers the target SOC distribution under the kernel terminal penalty and reduces the peak and time-averaged aggregate fleet demand under the running interaction cost.

The remainder of this paper is organized as follows. Section 2 recalls basic facts on the MMD and derives the Fourier representation used by both the terminal MMD penalty and the kernel congestion cost. Section 3 presents the proposed unbiased $O(NM)$ U -statistic estimators and establishes their unbiasedness, complexity, and variance properties. Section 4 formulates the empirical potential MFG problem and states the sample-level convergence theorem and the explicit almost-sure rate, together with the Schrödinger-bridge specialization. Section 5 presents numerical experiments, beginning with the Schrödinger-bridge special case as a warm-up and proceeding to the EV charging fleet coordination problem. Section 6 gives proofs of the main results.

Notation. Throughout the paper, $d \geq 1$ is a positive integer, and $|\cdot|$ denotes the Euclidean norm on $\mathbb{R}^d$. For any column $x \in \mathbb{R}^d$, we write $x^\top$ for the transposition of x and x_k for the k -th component of x , $k = 1, \dots, d$. For an $\mathbb{R}^d$ -valued random variable X , $\mathbb{E}[X]_k$ denotes the k -th component of its expectation vector $\mathbb{E}[X] \in \mathbb{R}^d$. The set of Borel probability measures on $\mathbb{R}^d$ is denoted by $\mathcal{P}(\mathbb{R}^d)$, and we write $\mathcal{L}(X) \in \mathcal{P}(\mathbb{R}^d)$ for the distribution of an $\mathbb{R}^d$ -valued random variable X . For a topological space E , $C_b(E)$ denotes the space of bounded continuous real-valued functions on E . For an integrable function $\Phi : \mathbb{R}^d \rightarrow \mathbb{R}$, $\hat{\Phi}$ denotes its Fourier transform with the convention $\hat{\Phi}(\xi) = (2\pi)^{-d/2} \int_{\mathbb{R}^d} e^{-i\xi^\top x} \Phi(x) dx$, where $i = \sqrt{-1}$. I_k denotes the $k \times k$ identity matrix, and we write $\mathcal{N}(m, \Sigma)$ for the Gaussian distribution on $\mathbb{R}^k$ with mean $m \in \mathbb{R}^k$ and covariance matrix Σ . $\mathbb{E}[\cdot]$ and $\text{Var}(\cdot)$ stand for expectation and variance, respectively. For probability measures Q, P on a common measurable space, $H(Q|P) := \int \log(dQ/dP) dQ$ when $Q \ll P$, and $H(Q|P) := +\infty$ otherwise, is the Kullback–Leibler divergence (relative entropy) of Q with respect to P . Finally, $C([0, 1]; \mathbb{R}^d)$ denotes the space of continuous paths $[0, 1] \rightarrow \mathbb{R}^d$ endowed with the supremum norm.

2 Hilbert space embeddings and Fourier representation of MMD

2.1 Maximum mean discrepancy

Let $K \in C_b(\mathbb{R}^d \times \mathbb{R}^d)$ be a symmetric and strictly positive definite kernel on $\mathbb{R}^d$, i.e., $K(x, y) = K(y, x)$ for $x, y \in \mathbb{R}^d$ and for any pairwise distinct $x_1, \dots, x_L \in \mathbb{R}^d$ and $\alpha = (\alpha_1, \dots, \alpha_L)^\top \in \mathbb{R}^L \setminus \{0\}$,

$$\sum_{j, \ell=1}^L \alpha_j \alpha_\ell K(x_j, x_\ell) > 0.$$

Then there exists a unique Hilbert space $\mathcal{H} \subset C_b(\mathbb{R}^d)$ such that K is a reproducing kernel on $\mathcal{H}$ with norm $\|\cdot\|$ (see, e.g., Wendland [36]); the inclusion $\mathcal{H} \subset C_b(\mathbb{R}^d)$ follows from the reproducing property

$|f(x)| = |\langle f, K(\cdot, x) \rangle| \leq \|f\| \sqrt{K(x, x)} \leq \|f\| \sup_z \sqrt{K(z, z)}$, which is finite since $K \in C_b(\mathbb{R}^d \times \mathbb{R}^d)$. The *maximum mean discrepancy* (MMD) associated with K is defined by

$$(2.1) \quad \gamma_K(\mu, \nu) := \sup_{f \in \mathcal{H}, \|f\| \leq 1} \left| \int_{\mathbb{R}^d} f d\mu - \int_{\mathbb{R}^d} f d\nu \right|, \quad \mu, \nu \in \mathcal{P}(\mathbb{R}^d)$$

(see Gretton et al. [14]). We assume that γ_K defines a metric on $\mathcal{P}(\mathbb{R}^d)$, i.e., K is a *characteristic kernel*. By the reproducing property,

$$(2.2) \quad \gamma_K(\mu, \nu)^2 = \iint_{\mathbb{R}^d \times \mathbb{R}^d} K(x, y) (\mu - \nu)(dx) (\mu - \nu)(dy)$$

(see Section 2 in Sriperumbudur et al. [32]).

2.2 Fourier representation

We now derive the Fourier representation that is central to our approach. Consider the case where

- (A) K is represented as $K(x, y) = \Phi(x - y)$, $x, y \in \mathbb{R}^d$, for some bounded, continuous and integrable function Φ on $\mathbb{R}^d$ such that its Fourier transform $\hat{\Phi}$ is also integrable on $\mathbb{R}^d$ and satisfies $\hat{\Phi}(\xi) > 0$ for any $\xi \in \mathbb{R}^d$.

Under (A), the kernel K is in $C_b(\mathbb{R}^d \times \mathbb{R}^d)$ that is symmetric and strictly positive definite. Moreover, the Fourier inversion formula gives

$$(2.3) \quad \Phi(x) = \int_{\mathbb{R}^d} \rho(\xi) e^{i\xi^\top x} d\xi, \quad x \in \mathbb{R}^d,$$

where $\rho(\xi) = (2\pi)^{-d/2} \hat{\Phi}(\xi) > 0$ for $\xi \in \mathbb{R}^d$. Substituting (2.3) into (2.2) and applying Fubini's theorem, we obtain

$$(2.4) \quad \gamma_K(\mu, \nu)^2 = \int_{\mathbb{R}^d} |\tilde{\mu}(\xi) - \tilde{\nu}(\xi)|^2 \rho(\xi) d\xi,$$

where $\tilde{\mu}(\xi) := \int_{\mathbb{R}^d} e^{i\xi^\top x} \mu(dx)$ denotes the characteristic function of μ .

Note that ρ integrates to $\Phi(0) = K(0, 0)$. Let $X \sim \mu$, $Y \sim \nu$, $X' \sim \mu$, and $Y' \sim \nu$, with X, X', Y, Y' mutually independent. Then, since $|\tilde{\mu}(\xi) - \tilde{\nu}(\xi)|^2$ is real-valued,

$$\begin{aligned} |\tilde{\mu}(\xi) - \tilde{\nu}(\xi)|^2 &= \mathbb{E}[e^{i\xi^\top X} - e^{i\xi^\top Y}] \overline{\mathbb{E}[e^{i\xi^\top X} - e^{i\xi^\top Y}]} \\ &= \mathbb{E} \left[e^{i\xi^\top (X - X')} - e^{i\xi^\top (X - Y')} - e^{i\xi^\top (Y - X')} + e^{i\xi^\top (Y - Y')} \right] \\ (2.5) \quad &= \mathbb{E}[\cos(\xi^\top (X - X'))] - 2\mathbb{E}[\cos(\xi^\top (X - Y))] + \mathbb{E}[\cos(\xi^\top (Y - Y'))], \end{aligned}$$

where in the last step we used the fact that (X, Y') and (X, Y) have the same joint distribution $\mu \otimes \nu$, and similarly $\mathbb{E}[\cos(\xi^\top (Y - X'))] = \mathbb{E}[\cos(\xi^\top (X - Y))]$ by the symmetry of $\cos$. Now, let Z be a random variable with density $\rho/\Phi(0)$, independent of (X, X', Y, Y') . Integrating (2.5) against $\rho(\xi) d\xi$ and using (2.4), we arrive at the representation

$$(2.6) \quad \gamma_K(\mu, \nu)^2 = \Phi(0) \mathbb{E} \left[\cos(Z^\top (X - X')) - 2\cos(Z^\top (X - Y)) + \cos(Z^\top (Y - Y')) \right].$$

Remark 2.1. The representation (2.6) also shows that γ_K metrizes the weak topology on $\mathcal{P}(\mathbb{R}^d)$ under (A). See [24, Proposition 2.1] for a proof.

Example 2.2. The Gaussian kernel $K(x, y) = e^{-\alpha|x-y|^2}$, $\alpha > 0$, satisfies assumption (A). Indeed, $K(x, y) = \Phi(x - y)$ with $\Phi(z) = e^{-\alpha|z|^2}$ bounded, continuous, and integrable on $\mathbb{R}^d$, and its inverse Fourier transform $\rho(\xi) = (2\pi)^{-d/2}\hat{\Phi}(\xi) = (4\alpha)^{-d/2}e^{-|\xi|^2/(4\alpha)} > 0$ is the density of $\mathcal{N}(0, 2\alpha I_d)/\Phi(0)$ up to the normalising constant $\Phi(0) = 1$. Hence $\Phi(0) = 1$ and the random frequency in (2.6) satisfies $Z \sim \mathcal{N}(0, 2\alpha I_d)$.

3 Unbiased estimator via random Fourier U -statistics

Throughout the remainder of the paper we fix a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$ on which all random variables are defined, and we denote by $\mathbb{E}$ the corresponding expectation.

3.1 Construction of the estimator

The representation (2.6) suggests a Monte Carlo approach: draw $Z_1, \dots, Z_M$ i.i.d. from the density $\rho/\Phi(0)$ and approximate γ_K^2 by averaging over the random frequencies. Given i.i.d. samples $X_1, \dots, X_N \sim \mu$ and $Y_1, \dots, Y_N \sim \nu$, we define for each $z \in \mathbb{R}^d$ the following quantities:

$$(3.1) \quad S_c^X(z) := \sum_{i=1}^N \cos(z^\top X_i), \quad S_s^X(z) := \sum_{i=1}^N \sin(z^\top X_i),$$

and similarly $S_c^Y(z)$, $S_s^Y(z)$ for the Y -samples.

The key observation is that the U -statistic for $\mathbb{E}[\cos(z^\top(X - X'))]$ can be computed in $O(N)$:

$$(3.2) \quad U_{XX}(z) := \frac{(S_c^X(z))^2 + (S_s^X(z))^2 - N}{N(N-1)}.$$

Indeed, expanding the squares and using the trigonometric identity $\cos(a - b) = \cos a \cos b + \sin a \sin b$, we get

$$\begin{aligned} (S_c^X)^2 + (S_s^X)^2 &= \sum_{i,j=1}^N (\cos(z^\top X_i) \cos(z^\top X_j) + \sin(z^\top X_i) \sin(z^\top X_j)) \\ &= \sum_{i,j=1}^N \cos(z^\top (X_i - X_j)) = N + \sum_{i \neq j} \cos(z^\top (X_i - X_j)). \end{aligned}$$

Dividing by $N(N-1)$ yields an unbiased estimator of $\mathbb{E}[\cos(z^\top(X - X'))]$.

Similarly, the cross term $\mathbb{E}[\cos(z^\top(X - Y))]$ is estimated by

$$(3.3) \quad V_{XY}(z) := \frac{S_c^X(z) S_c^Y(z) + S_s^X(z) S_s^Y(z)}{N^2},$$

which is unbiased since X_i and Y_j are always from independent samples.

Combining the representation (2.6) with the Monte Carlo approach, we propose the *RF U-statistic estimator*: given i.i.d. random frequencies $Z_1, \dots, Z_M \sim \rho/\Phi(0)$ independent of the data, set

$$(3.4) \quad \hat{\gamma}_{M,N}^2 := \frac{\Phi(0)}{M} \sum_{r=1}^M [U_{XX}(Z_r) - 2V_{XY}(Z_r) + U_{YY}(Z_r)].$$

The following theorem collects the basic properties of our penalty term. The proof is given in Section 6.1.

Theorem 3.1 (Unbiasedness, complexity, and variance). *Suppose (A) holds. Let $X_1, \dots, X_N$ with $N \geq 2$, $Y_1, \dots, Y_N$, and $Z_1, \dots, Z_M$ with $M \geq 1$ be three mutually independent collections of i.i.d. random variables with $X_i \sim \mu$, $Y_j \sim \nu$, and $Z_r \sim \rho/\Phi(0)$. Then the following hold.*

- (i) $\mathbb{E}[\hat{\gamma}_{M,N}^2] = \gamma_K(\mu, \nu)^2$.
- (ii) The computational complexity of $\hat{\gamma}_{M,N}^2$ is $O(NM)$.
- (iii) $\text{Var}(\hat{\gamma}_{M,N}^2) = O(1/M) + O(1/N)$.

Remark 3.2. The standard RFF approach computes a V -statistic $|\frac{1}{N} \sum_i \phi(X_i) - \frac{1}{N} \sum_j \phi(Y_j)|^2$ with positive bias of order $\Phi(0)/N$ when $\mu = \nu$. Our $\hat{\gamma}_{M,N}^2$ eliminates this bias while preserving the $O(NM)$ complexity, which avoids a non-vanishing penalty floor at the population level and yields the clean variance decomposition of Theorem 3.1(iii) and the explicit a.s. rate of Proposition 4.9 below without any bias-correction term. Empirically the bias is mild near the optimum. Section 5.1.4 shows that the trained drifts from the biased V -statistic and the unbiased U -statistic are indistinguishable across the tested range of (N, λ) . The unbiased construction is thus preferred for theoretical reasons rather than for an immediate empirical accuracy gain.

3.2 Estimator for the kernel interaction cost

The same Fourier U -statistic construction extends to the running interaction cost $\mathcal{R}[\nu]$ defined by (4.3). Suppose (A) holds for the congestion kernel $W(x, y) = \Psi(x - y)$, with generating density $\rho_W(\xi) := (2\pi)^{-d/2} \hat{\Psi}(\xi) > 0$. The Fourier inversion formula and Fubini's theorem (cf. Section 2.2) then give

$$(3.5) \quad \iint_{\mathbb{R}^d \times \mathbb{R}^d} W(x, y) \nu(dx) \nu(dy) = \Psi(0) \mathbb{E}[\cos(\tilde{Z}^\top (X - X'))],$$

where $\tilde{Z} \sim \rho_W/\Psi(0)$ and X, X' is an independent pair of samples from ν , all mutually independent. Given i.i.d. samples $X_1, \dots, X_N \sim \nu$ ($N \geq 2$) and i.i.d. random frequencies $\tilde{Z}_1, \dots, \tilde{Z}_M \sim \rho_W/\Psi(0)$ ($M \geq 1$), independent of the data, define

$$(3.6) \quad \hat{\mathcal{R}}_{M,N}[\hat{\nu}^N] := \frac{\Psi(0)}{2M} \sum_{r=1}^M U_{XX}(\tilde{Z}_r) = \frac{\Psi(0)}{2M} \sum_{r=1}^M \frac{S_c^X(\tilde{Z}_r)^2 + S_s^X(\tilde{Z}_r)^2 - N}{N(N-1)},$$

where U_{XX} , S_c^X , S_s^X are as in (3.1)–(3.2) evaluated at the single sample collection $\{X_i\}_{i=1}^N$. The estimator (3.6) is the natural analogue of $\hat{\gamma}_{M,N}^2$ for the self-interaction kernel functional $\frac{1}{2} \iint W d\nu \otimes d\nu$: it removes the diagonal bias of the V -statistic via the same trigonometric identity and is also evaluable in $O(NM)$.

Theorem 3.3 (Properties of the interaction estimator). *Suppose (A) holds for W . Let $X_1, \dots, X_N$ with $N \geq 2$ and $\tilde{Z}_1, \dots, \tilde{Z}_M$ with $M \geq 1$ be two mutually independent collections of i.i.d. random variables with $X_i \sim \nu$ and $\tilde{Z}_r \sim \rho_W / \Psi(0)$. Then*

- (i) $\mathbb{E}[\hat{\mathcal{R}}_{M,N}[\hat{\nu}^N]] = \mathcal{R}[\nu]$.
- (ii) The computational complexity of $\hat{\mathcal{R}}_{M,N}[\hat{\nu}^N]$ is $O(NM)$.
- (iii) $\text{Var}(\hat{\mathcal{R}}_{M,N}[\hat{\nu}^N]) = O(1/M) + O(1/N)$.

Proof. The proof is parallel to that of Theorem 3.1 and is given in Section 6.2. $\square$

4 Potential mean-field game formulation and convergence analysis

4.1 Potential MFG with kernel costs

Fix the horizon $T > 0$, the diffusion coefficient $\sigma > 0$, the initial law $\mu_0 \in \mathcal{P}(\mathbb{R}^d)$ with finite second moment, and the *deadline target* $\mu_T \in \mathcal{P}(\mathbb{R}^d)$ with finite second moment. Here μ_T prescribes the distribution that the controlled population is required to attain at the terminal time $t = T$, and the deviation $\mathcal{L}(X_T)$ from μ_T is enforced softly via a kernel-MMD penalty (introduced in (4.2) below). For instance, in the EV charging fleet of Section 5.6, μ_T is the desired state-of-charge distribution of the fleet at the end of the operational horizon. We fix a reproducing kernel $K : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ satisfying assumption (A) with generating function Φ , the terminal penalty weight $\lambda > 0$, and the congestion weight $c \geq 0$. The running mean-field cost $\mathcal{R} : \mathcal{P}(\mathbb{R}^d) \rightarrow [0, \infty)$ is a non-negative weakly continuous functional, specified later either as a self-interaction integral with a kernel W satisfying (A) (Section 3.2 and (4.3) below), or as the aggregate-demand cost used in the EV application of Section 5.6.

We endow the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ of Section 3 with a filtration $\{\mathcal{F}_t\}_{t \in [0, T]}$ satisfying the usual conditions, and assume that it supports a d -dimensional $\{\mathcal{F}_t\}$ -Brownian motion B and an $\mathcal{F}_0$ -measurable random variable $X_0 \sim \mu_0$ independent of B . The controlled state equation is

$$(4.1) \quad dX_t = u(t, X_t) dt + \sigma dB_t, \quad X_0 \sim \mu_0, \quad t \in [0, T].$$

We define the class of *admissible controls* $\mathcal{A}$ as the set of Borel measurable functions $u : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ for which (4.1) admits a unique strong solution X satisfying $\mathbb{E}[\int_0^T |u(t, X_t)|^2 dt] < \infty$. For example, any Borel function u that is bounded and Lipschitz in x (uniformly in t) belongs to $\mathcal{A}$.

The *potential mean-field game* we consider is

$$(4.2) \quad H_{\lambda, c}^* := \inf_{u \in \mathcal{A}} \left\{ \frac{1}{2} \mathbb{E} \left[\int_0^T |u(t, X_t)|^2 dt \right] + c \int_0^T \mathcal{R}[\mathcal{L}(X_t)] dt + \lambda \mathcal{T}[\mathcal{L}(X_T)] \right\},$$

where the kernel-based interaction and terminal costs are

$$(4.3) \quad \mathcal{R}[\nu] = \frac{1}{2} \iint_{\mathbb{R}^d \times \mathbb{R}^d} W(x, y) \nu(dx) \nu(dy), \quad \mathcal{T}[\nu] = \gamma_K(\nu, \mu_T)^2.$$

The interaction $\mathcal{R}$ is a quadratic functional of the population law and admits the variational derivative $\delta\mathcal{R}[\nu]/\delta\nu(x) = \int W(x, y) \nu(dy)$, making (4.2) a potential MFG in the sense of Lasry–Lions [17]; the terminal cost $\mathcal{T}[\nu]$ has variational derivative $\delta\mathcal{T}[\nu]/\delta\nu(x) = 2(\int K(x, y)\nu(dy) - \int K(x, y)\mu_T(dy))$ and equals zero exactly when $\nu = \mu_T$.

Remark 4.1 (Behaviour of $\mathcal{R}$). Under (A), the Bochner representation gives $\mathcal{R}[\nu] = \frac{1}{2} \int |\hat{\nu}(\xi)|^2 \rho_W(\xi) d\xi$, so $\mathcal{R}$ takes values in $[0, \Psi(0)/2]$, is translation invariant, and is monotone in the concentration of ν . The maximum $\Psi(0)/2$ is attained at Dirac masses and $\mathcal{R}[\nu] \rightarrow 0$ as ν spreads. For the Gaussian $W(x, y) = e^{-\alpha|x-y|^2}$ and $\nu = \mathcal{N}(m, \sigma^2 I_d)$, $\mathcal{R}[\nu] = \frac{1}{2}(1 + 4\alpha\sigma^2)^{-d/2}$. The bandwidth α thus controls the scale at which population concentration is penalized, and $c \int_0^T \mathcal{R}[\mathcal{L}(X_t)] dt$ incentivizes the optimal drift to spread agents apart on the scale $\alpha^{-1/2}$.

At each iteration the cost in (4.2) is estimated from N i.i.d. controlled-path samples $X^{(1)}, \dots, X^{(N)}$ of (4.1) under the candidate u , N i.i.d. target samples $Y^{(1)}, \dots, Y^{(N)} \sim \mu_T$, and M random frequencies for each kernel (all jointly independent; see Algorithm 4.1 for the precise schedule). Writing $\hat{\nu}_t^N := \frac{1}{N} \sum_{i=1}^N \delta_{X_t^{(i)}}$ for the empirical distribution at time t , the empirical kernel costs are

$$(4.4) \quad \hat{\mathcal{R}}_{M,N}[\hat{\nu}_t^N] := \frac{\Psi(0)}{M} \sum_{r=1}^M \frac{S_c^{X_t}(\tilde{Z}_r)^2 + S_s^{X_t}(\tilde{Z}_r)^2 - N}{2N(N-1)},$$

$$(4.5) \quad \hat{\mathcal{T}}_{M,N}[\hat{\nu}_T^N, \mu_T] := \hat{\gamma}_{M,N}^2(\hat{\nu}_T^N, \mu_T),$$

where $\hat{\gamma}_{M,N}^2$ is the RF U -statistic estimator of Section 3 applied to the terminal samples, and $S_c^{X_t}, S_s^{X_t}$ are the trigonometric sums of (3.1) evaluated at the N controlled paths at time t . The empirical mean-field game problem is

$$(4.6) \quad \hat{H}_{\lambda,c,M,N} := \inf_{u \in \mathcal{A}} \left\{ \frac{1}{2N} \sum_{i=1}^N \int_0^T |u(t, X_t^{(i)})|^2 dt + c \int_0^T \hat{\mathcal{R}}_{M,N}[\hat{\nu}_t^N] dt + \lambda \hat{\mathcal{T}}_{M,N}[\hat{\nu}_T^N, \mu_T] \right\}.$$

Note that the infimum is taken over deterministic feedback controls $u \in \mathcal{A}$. The objective is a random functional of u through the four sample collections, and at each gradient step Algorithm 4.1 (Section 4.3) draws a fresh independent realization (see Remark 4.16).

Remark 4.2 (Schrödinger-bridge specialization). Setting $c = 0$ and $T = 1$ in (4.2) reduces the potential MFG to the kernel-MMD-penalty Schrödinger bridge of [24], namely

$$(4.7) \quad H_{\lambda,0}^*|_{T=1} = \inf_{u \in \mathcal{A}} \left\{ \frac{1}{2} \mathbb{E} \left[\int_0^1 |u(t, X_t)|^2 dt \right] + \lambda \gamma_K(\mathcal{L}(X_1), \mu_1)^2 \right\}.$$

The sample-level convergence theorem for this special case is given as Corollary 4.7 below.

Remark 4.3. The classical mean-field game and Schrödinger bridge problems are often formulated in the *weak* sense, where the probability space and the driving Brownian motion are themselves part

of the optimization (see, e.g., [19]). Since the conclusions of our convergence analysis (Theorem 4.6 below) depend only on the laws of the controlled processes and on the empirical value $\hat{H}_{\lambda,c,M,N}$, both of which are invariant under the choice of the underlying probability space, the strong-solution formulation based on $\mathcal{A}$ adopted here is sufficient and matches the implementation in Algorithm 4.1, where u is parametrized as a neural network feedback.

4.2 Sample-level convergence

Before stating the main theorem, we record the strong consistency of the empirical objective, which underlies the convergence analysis.

Lemma 4.4 (Strong consistency of the MMD estimator). *Suppose that (A) holds. Let $\{M_n\}_{n \geq 1}$ and $\{N_n\}_{n \geq 1}$ be deterministic sequences with $N_n \rightarrow \infty$ and $M_n/\log n \rightarrow \infty$. Then, for any $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$,*

$$\hat{\gamma}_{M_n, N_n}^2 \longrightarrow \gamma_K(\mu, \nu)^2 \quad n \rightarrow \infty, \quad a.s.$$

Proof. See Section 6.3. □

Kolmogorov's strong law of large numbers immediately yields the following:

Lemma 4.5 (Strong consistency of the empirical energy). *Fix a deterministic feedback control $u \in \mathcal{A}$. Let $(B^{(i)}, X_0^{(i)})_{i \geq 1}$ be a sequence of independent copies of (B, X_0) (with B, X_0 , and μ_0 as in (4.1)), and let $X^{(i)}$ denote the unique strong solution of (4.1) driven by $(B^{(i)}, X_0^{(i)})$ under the control u . Then the $X^{(i)}$ are i.i.d. copies of the controlled process X , and*

$$\frac{1}{N} \sum_{i=1}^N \int_0^1 |u(t, X_t^{(i)})|^2 dt \longrightarrow \mathbb{E} \left[\int_0^1 |u(t, X_t)|^2 dt \right], \quad N \rightarrow \infty, \quad a.s.$$

We are now in a position to state the main convergence result for the empirical mean-field game problem (4.6). Put

$$H_c^* = \inf_{u \in \mathcal{A}} \left\{ \frac{1}{2} \mathbb{E} \left[\int_0^T |u(t, X_t)|^2 dt \right] + c \int_0^T \mathcal{R}[\mathcal{L}(X_t)] dt : \mathcal{L}(X_T) = \mu_T \right\}.$$

Theorem 4.6 (Convergence for the potential MFG). *Suppose that μ_0 and μ_T have finite second moments and that (A) holds for the kernels K and W . Fix the congestion weight $c \geq 0$. Let $\{\lambda_n\}$, $\{M_n\}$, $\{N_n\}$, and $\{\varepsilon_n\}$ be deterministic sequences satisfying $\lambda_n, \varepsilon_n > 0$, $M_n, N_n \in \mathbb{N}$, and*

$$(4.8) \quad \lambda_n \rightarrow \infty, \quad N_n \rightarrow \infty, \quad M_n/\log n \rightarrow \infty, \quad \varepsilon_n \rightarrow 0, \quad \lambda_n \varepsilon_n \rightarrow 0, \quad n \rightarrow \infty.$$

For each n , let $u_n \in \mathcal{A}$ be an ε_n -optimal admissible control for the empirical MFG (4.6) with $\lambda = \lambda_n$, c fixed, $M = M_n$, $N = N_n$, and let $X^{(n)}$ denote the controlled process generated under u_n on i.i.d. sample paths $X^{(n,1)}, \dots, X^{(n,N_n)}$, target i.i.d. samples $Y^{(n,1)}, \dots, Y^{(n,N_n)} \sim \mu_T$, and i.i.d. random frequencies $Z_1^{(n)}, \dots, Z_{M_n}^{(n)} \sim \rho_K/\Phi(0)$ and $\tilde{Z}_1^{(n)}, \dots, \tilde{Z}_{M_n}^{(n)} \sim \rho_W/\Psi(0)$, all drawn independently of the data used to construct u_n . Then, almost surely,

$$(i) \quad \lim_{n \rightarrow \infty} \sqrt{\lambda_n} \gamma_K(\mathcal{L}(X_T^{(n)}), \mu_T) = 0,$$

$$(ii) \lim_{n \rightarrow \infty} \hat{H}_{\lambda_n, c, M_n, N_n} = H_c^*.$$

Proof. See Section 6.4. □

Corollary 4.7 (Schrödinger-bridge specialization). *Suppose that $c = 0$ and $T = 1$. Under the same conditions (4.8) but with (A) required only for K , the ε_n -optimal control u_n of the empirical SBP penalty problem satisfies, almost surely,*

$$(i) \lim_{n \rightarrow \infty} \sqrt{\lambda_n} \gamma_K(\mathcal{L}(X_1^{(n)}), \mu_1) = 0,$$

$$(ii) \lim_{n \rightarrow \infty} \hat{H}_{\lambda_n, 0, M_n, N_n} = H^*,$$

where H^* denotes the Schrödinger-bridge value

$$H^* := H_c^*|_{c=0, T=1} = \inf_{u \in \mathcal{A}} \left\{ \frac{1}{2} \mathbb{E} \left[\int_0^1 |u(t, X_t)|^2 dt \right] : \mathcal{L}(X_1) = \mu_1 \right\}.$$

Proposition 4.8 (Convergence with a general running cost). *Let $\mathcal{R} : \mathcal{P}(\mathbb{R}^d) \rightarrow [0, \infty)$ be a non-negative functional that is continuous in the weak topology of $\mathcal{P}(\mathbb{R}^d)$, and, for each $N \geq 2$, let $\hat{\mathcal{R}}_N : (\mathbb{R}^d)^N \rightarrow [0, \infty)$ be a Borel measurable symmetric function; for $\nu \in \mathcal{P}(\mathbb{R}^d)$ and i.i.d. random variables $\xi_1, \dots, \xi_N \sim \nu$, write $\hat{\mathcal{R}}_N[\hat{\nu}^N] := \hat{\mathcal{R}}_N(\xi_1, \dots, \xi_N)$ where $\hat{\nu}^N := N^{-1} \sum_{i=1}^N \delta_{\xi_i}$. Assume that $\hat{\mathcal{R}}_N$ is strongly consistent: for every $\nu \in \mathcal{P}(\mathbb{R}^d)$ and every deterministic sequence $\{N_n\}$ with $N_n \rightarrow \infty$,*

$$(4.9) \quad \hat{\mathcal{R}}_{N_n}[\hat{\nu}^{N_n}] \xrightarrow{\text{a.s.}} \mathcal{R}[\nu] \quad (n \rightarrow \infty).$$

Define the modified empirical MFG by replacing $c\hat{\mathcal{R}}_{M,N}[\hat{\nu}_t^N]$ in (4.6) by $c\hat{\mathcal{R}}_N[\hat{\nu}_t^N]$, and denote the resulting value by $\hat{H}_{\lambda, c, M, N}^{(\mathcal{R})}$.

Suppose that μ_0 and μ_T have finite second moments, that (A) holds for the terminal kernel K , that $\{\lambda_n\}, \{M_n\}, \{N_n\}, \{\varepsilon_n\}$ are deterministic sequences satisfying $\lambda_n, \varepsilon_n > 0, M_n, N_n \in \mathbb{N}$, and (4.8), and that for each $n \geq 1$, $u_n \in \mathcal{A}$ is an ε_n -optimal admissible control for $\hat{H}_{\lambda_n, c, M_n, N_n}^{(\mathcal{R})}$. Let $X^{(n)}$ denote the controlled process generated under u_n on fresh sample paths $X^{(n,1)}, \dots, X^{(n, N_n)}$, target i.i.d. samples $Y^{(n,1)}, \dots, Y^{(n, N_n)} \sim \mu_T$, and i.i.d. random frequencies $Z_1^{(n)}, \dots, Z_{M_n}^{(n)} \sim \rho_K / \Phi(0)$, all drawn independently of the data used to construct u_n . Then, almost surely,

$$(i) \lim_{n \rightarrow \infty} \sqrt{\lambda_n} \gamma_K(\mathcal{L}(X_T^{(n)}), \mu_T) = 0,$$

$$(ii) \lim_{n \rightarrow \infty} \hat{H}_{\lambda_n, c, M_n, N_n}^{(\mathcal{R})} = H_c^*,$$

where

$$H_c^* := \inf_{u \in \mathcal{A}} \left\{ \frac{1}{2} \mathbb{E} \left[\int_0^T |u(t, X_t)|^2 dt \right] + c \int_0^T \mathcal{R}[\mathcal{L}(X_t)] dt : \mathcal{L}(X_T) = \mu_T \right\}.$$

Proof. See Section 6.5. □

The condition $\lambda_n \varepsilon_n \rightarrow 0$ requires the optimization tolerance ε_n to vanish faster than the inverse penalty $1/\lambda_n$. The rate condition $M_n/\log n \rightarrow \infty$ is mild and is automatically satisfied by any practical choice such as M_n polynomial in n . It is used in Lemma 4.4 to ensure that the random fluctuation $\hat{\gamma}_{M_n, N_n}^2 - \gamma_K^2$ vanishes almost surely, which in turn underlies the almost-sure convergence claimed below. Combined with $\lambda_n \varepsilon_n \rightarrow 0$, this ensures that the penalty fluctuation is dominated by the penalty growth.

Theorem 4.6 is established at the *sample* level, directly for the empirical objective (4.6) minimized by Algorithm 4.1, in contrast to the population-level analyses of [24, 29], which work with the exact γ_K^2 and the expected energy. This reduction is enabled by Lemmas 4.4 and 4.5.

The almost-sure convergence in Theorem 4.6 (i) can be sharpened into an explicit rate by upgrading Lemma 4.4 to its quantitative Hoeffding form, making the trade-off between the penalty schedule and the sample/feature counts explicit.

Proposition 4.9 (Rate of convergence, SBP case). *Suppose the hypotheses of Corollary 4.7 hold. Suppose moreover that the following hold:*

- (H1) *There exist $L \geq 0$ and an optimal control $u^* \in \mathcal{A}$ for (4.7) such that $\|u^*\|_\infty \leq L$ and $\|u_n\|_\infty \leq L$ for every $n \geq 1$, where $\|u\|_\infty := \sup_{(t,x) \in [0,1] \times \mathbb{R}^d} |u(t,x)|$.*
- (H2) *At each $n \geq 1$, the samples $\{X^{(n,i)}\}_{i=1}^{N_n}$, $\{Y^{(n,j)}\}_{j=1}^{N_n}$, $\{Z_r^{(n)}\}_{r=1}^{M_n}$ appearing in (4.6) are independent of the random elements used to construct u_n .*

Then there exists a constant $C^ > 0$, depending only on $\Phi(0)$ and L , such that, for all sufficiently large n , almost surely, the observable empirical penalty satisfies*

$$(4.10) \quad \hat{\gamma}_{M_n, N_n}^2(\mathcal{L}(X_1^{(n)}), \mu_1) \leq \frac{H^* + \varepsilon_n}{\lambda_n} + C^* \left(\sqrt{\frac{\log n}{M_n}} + \sqrt{\frac{\log n}{N_n}} \right),$$

and consequently the population-level discrepancy satisfies

$$(4.11) \quad \gamma_K(\mathcal{L}(X_1^{(n)}), \mu_1)^2 \leq \frac{H^* + \varepsilon_n}{\lambda_n} + 2C^* \left(\sqrt{\frac{\log n}{M_n}} + \sqrt{\frac{\log n}{N_n}} \right).$$

In particular, with the polynomial schedule $\lambda_n = n^a$, $M_n = N_n = n^b$, $\varepsilon_n = n^{-e}$ ($a, b, e > 0$ and $e \geq a$), the right-hand sides of (4.10)–(4.11) are both $O(n^{-\min(a, b/2)}(\log n)^{1/2})$, and the optimal balance $b = 2a$ gives

$$\gamma_K(\mathcal{L}(X_1^{(n)}), \mu_1) = O\left(n^{-a/2}(\log n)^{1/4}\right) \quad a.s.$$

Proof. See Section 6.6. □

Remark 4.10. The two bounds in Proposition 4.9 are complementary: (4.10) is verifiable from data (computed on a fresh evaluation batch in Section 5), while (4.11) is the theoretical guarantee on the unobservable distributional distance $\gamma_K(\mathcal{L}(X_1^{(n)}), \mu_1)$. Both decompose the error into the penalty bias H^*/λ_n , the optimization tolerance ε_n/λ_n , and the sample/feature concentration $\sqrt{\log n/M_n} + \sqrt{\log n/N_n}$ (the latter reduces to $\sqrt{\log n/N_n}$ in the kernel U -statistic limit of Corollary 4.12). Hypothesis (H1) is automatic for drift networks with bounded output activations or weight clipping, and can be relaxed to a uniform L^4 -moment bound at the cost of using Bernstein's inequality; (H2) holds whenever (4.6) is evaluated on a fresh batch.

Proposition 4.11 (Rate of convergence for the potential MFG). *Suppose the hypotheses of Theorem 4.6 hold, together with the analogues of (H1)–(H2) for the MFG horizon $[0, T]$, namely $\|u^*\|_\infty \leq L$ and $\|u_n\|_\infty \leq L$ for every $n \geq 1$ with $\|u\|_\infty := \sup_{(t,x) \in [0,T] \times \mathbb{R}^d} |u(t,x)|$, and at each $n \geq 1$ the samples $\{X^{(n,i)}\}_{i=1}^{N_n}$, $\{Y^{(n,j)}\}_{j=1}^{N_n}$, $\{Z_r^{(n)}\}_{r=1}^{M_n}$, $\{\tilde{Z}_r^{(n)}\}_{r=1}^{M_n}$ appearing in (4.6) are independent of the random elements used to construct u_n . Then there exists a constant $C_c^* > 0$, depending on $\Phi(0), \Psi(0), L, c, T$ only, such that*

$$(4.12) \quad \gamma_K(\mathcal{L}(X_T^{(n)}), \mu_T)^2 \leq \frac{H_c^* + \varepsilon_n}{\lambda_n} + C_c^* \left(\sqrt{\frac{\log n}{M_n}} + \sqrt{\frac{\log n}{N_n}} \right)$$

for all sufficiently large n , almost surely; in particular, under the polynomial schedule of Proposition 4.9, $\gamma_K(\mathcal{L}(X_T^{(n)}), \mu_T) = O(n^{-a/2}(\log n)^{1/4})$ a.s.

Proof. See Section 6.7. □

The limit $M \rightarrow \infty$ in Theorem 4.6 corresponds to replacing the random Fourier U -statistic by the kernel U -statistic estimator $\bar{\gamma}_K^2(D)$ (cf. (1.3) and the proof of Theorem 3.1(iii) in Section 6.1), evaluated on the controlled-path terminal samples $X_1^{(1)}, \dots, X_1^{(N)}$ and target i.i.d. samples $Y^{(1)}, \dots, Y^{(N)} \sim \mu_T$:

$$(4.13) \quad \bar{\gamma}_K^2(D) = \frac{1}{N(N-1)} \sum_{i \neq j} K(X_1^{(i)}, X_1^{(j)}) - \frac{2}{N^2} \sum_{i,j} K(X_1^{(i)}, Y^{(j)}) + \frac{1}{N(N-1)} \sum_{i \neq j} K(Y^{(i)}, Y^{(j)}).$$

This is the population MMD² estimator used in [24]. The corresponding empirical penalty problem is

$$(4.14) \quad \hat{H}_{\lambda,N}^{\text{ker}} := \frac{1}{2} \inf_{u \in \mathcal{A}} \left\{ \frac{1}{N} \sum_{i=1}^N \int_0^1 |u(t, X_t^{(i)})|^2 dt + \lambda \bar{\gamma}_K^2(D) \right\}.$$

The almost-sure convergence of $\hat{H}_{\lambda,N}^{\text{ker}}$ to H^* then follows from Theorem 4.6, without any rate condition on the random-feature count. Formally, the kernel- U -statistic penalty problem (4.14) is the $M = \infty$ limit of the random Fourier penalty problem (4.6) in the sense that $\hat{\gamma}_{M,N}^2 \rightarrow \bar{\gamma}_K^2(D)$ a.s. as $M \rightarrow \infty$ for fixed N (by the conditional tower decomposition (6.3); see Remark 6.1), so the next corollary can be read as the limiting case of Theorem 4.6 with the random-feature ingredient replaced by its expectation:

Corollary 4.12. *Suppose that μ_0 and μ_1 have finite second moments and that (A) holds. Let $\{\lambda_n\}$, $\{N_n\}$, and $\{\varepsilon_n\}$ be sequences satisfying*

$$\lambda_n \rightarrow \infty, \quad N_n \rightarrow \infty, \quad \varepsilon_n \rightarrow 0, \quad \text{and} \quad \lambda_n \varepsilon_n \rightarrow 0.$$

For each n , let $u_n \in \mathcal{A}$ be an ε_n -optimal admissible control for the kernel U -statistic penalty problem (4.14) with $\lambda = \lambda_n$ and $N = N_n$, and let $X^{(n)}$ denote the corresponding controlled process. Then, almost surely,

$$(i) \quad \lim_{n \rightarrow \infty} \sqrt{\lambda_n} \gamma_K(\mathcal{L}(X_1^{(n)}), \mu_1) = 0,$$

$$(ii) \lim_{n \rightarrow \infty} \hat{H}_{\lambda_n, N_n}^{\text{ker}} = H^*.$$

Proof. See Section 6.8. □

Remark 4.13. Corollary 4.12 closes the sample-level gap left open by [24], whose original analysis works with the exact γ_K^2 and the expected energy even though the implemented algorithm uses their empirical counterparts. The same construction extends to the Monge case of [29]; see Remark 4.14.

Remark 4.14. An analogous convergence result holds for the Monge problem with $c(x, y) = |x - y|^2$ under the same conditions (4.8). In that case, the conclusions become: (i) $\sqrt{\lambda_n} \gamma_K(\mu \circ T_n^{-1}, \nu) \rightarrow 0$, and (ii) $\int |x - T_n(x)|^2 \mu(dx) \rightarrow \int |x - T^*(x)|^2 \mu(dx)$, where T^* is the optimal transport map. See [29, Theorem 2.1] for the population-level result and the analogue of Corollary 4.12 for its sample-based counterpart.

4.3 Algorithm

The empirical mean-field game problem (4.6) is implemented by parametrizing the drift u_θ of the controlled SDE (4.1) as a neural network and minimizing the empirical objective by stochastic gradient descent; see Algorithm 4.1. At each iteration the empirical objective is evaluated by combining the unbiased $O(NM)$ estimators of γ_K^2 (Theorem 3.1) and of $\mathcal{R}[\nu_t]$ (Theorem 3.3). The Schrödinger-bridge specialization $c = 0$ recovers the algorithm of [24] with the $O(N^2)$ kernel U -statistic replaced by the $O(NM)$ RF U -statistic.

Algorithm 4.1 Potential MFG with kernel costs via RF U -statistic estimators

Input: Number n of iterations, batch size N , random-feature count M , terminal penalty $\lambda > 0$, congestion weight $c \geq 0$, terminal kernel bandwidth α_K , congestion kernel bandwidth α_W , diffusion coefficient $\sigma > 0$, time horizon $T > 0$, time grid $0 = t_0 < t_1 < \dots < t_q = T$ with $q \in \mathbb{N}$ grid points.

Output: Neural-network drift u_θ

```

1: Initialize  $\theta$ 
2: for  $\ell = 1, 2, \dots, n$  do
3:    $\{Y_j\}_{j=1}^N \leftarrow$  i.i.d. samples from  $\mu_T$ 
4:    $\{Z_r\}_{r=1}^M \leftarrow$  i.i.d. samples from  $\rho_K/\Phi(0)$ ,  $\{\tilde{Z}_r\}_{r=1}^M \leftarrow$  i.i.d. from  $\rho_W/\Psi(0)$ 
5:   Simulate  $N$  independent paths  $\{X_t^{(i)}\}_{t \in [0, T]}$ ,  $i = 1, \dots, N$ , via Euler–Maruyama
6:    $\hat{\mathcal{C}}(\theta) \leftarrow \frac{1}{N} \sum_{i=1}^N \int_0^T |u_\theta(t, X_t^{(i)})|^2 dt$   $\triangleright$  empirical energy
7:    $\hat{\gamma}_{M, N}^2 \leftarrow$  (3.4) using  $\{X_T^{(i)}\}_{i=1}^N$ ,  $\{Y_j\}$ ,  $\{Z_r\}$   $\triangleright$  terminal MMD2 penalty
8:   if  $c > 0$  then
9:      $\hat{\mathcal{R}}_{\text{tot}}(\theta) \leftarrow \sum_{k=1}^q \hat{\mathcal{R}}_{M, N}[\hat{\nu}_{t_k}^N](t_k - t_{k-1})$   $\triangleright$  time-integrated empirical interaction via (3.6)
10:  else
11:     $\hat{\mathcal{R}}_{\text{tot}}(\theta) \leftarrow 0$ 
12:  end if
13:   $F(\theta) \leftarrow \frac{1}{\lambda} \hat{\mathcal{C}}(\theta) + \frac{c}{\lambda} \hat{\mathcal{R}}_{\text{tot}}(\theta) + \hat{\gamma}_{M, N}^2$   $\triangleright$  equivalent up to scaling to  $\hat{\mathcal{C}} + c \hat{\mathcal{R}}_{\text{tot}} + \lambda \hat{\gamma}_{M, N}^2$ 
14:  Take gradient step on  $\nabla_\theta F(\theta)$ 
15: end for
```

Remark 4.15. The interaction estimator (3.6) is computed on a *single* batch of N controlled-path samples, in contrast to the terminal estimator (3.4) which requires both controlled-path and target samples. In Algorithm 4.1, the same batch $\{X^{(i)}\}_{i=1}^N$ is used to evaluate $\hat{\mathcal{R}}_{M,N}[\hat{\nu}_{t_k}^N]$ at each time step t_k along the trajectory and to evaluate $\hat{\gamma}_{M,N}^2(\hat{\nu}_T^N, \mu_T)$ at the terminal time; the target samples $\{Y_j\}_{j=1}^N$ from μ_T and the two independent collections of random frequencies $\{Z_r\}$ for K and $\{\tilde{Z}_r\}$ for W enter only through the terminal and interaction penalties, respectively.

Remark 4.16. At each iteration, the random frequencies $Z_1, \dots, Z_M$, the path noise driving the N Euler–Maruyama trajectories, and the target samples $Y_1, \dots, Y_N$ are drawn independently and afresh. This “re-sampling” strategy ensures that, conditional on the current parameter θ , the stochastic gradient of $F(\theta)$ is an unbiased estimator of the population gradient (by Theorem 3.1(i)), and it provides the sample-independence used in the convergence analysis above (see hypothesis (H2) of Proposition 4.9). The U -statistic structure of $\hat{\gamma}_{M,N}^2$ does not require splitting the N paths into sub-batches: the diagonal-removing decomposition (3.2)–(3.3) already eliminates the biasing self-terms, so all N paths participate jointly as the X -sample of the estimator.

Remark 4.17. The same estimator applies to the Monge optimal transport problem studied in [29], as well as to its multi-marginal extension in [25]. In the Monge setting, the SDE is replaced by a deterministic transport map T_θ , and the objective becomes $\frac{1}{N} \sum_i c(X_i, T_\theta(X_i)) + \hat{\gamma}_{M,N}^2$ with $\hat{\gamma}_{M,N}^2$ computed from $\{T_\theta(X_i)\}$ and target samples $\{Y_j\}$.

5 Numerical experiments

All numerical experiments are implemented in PyTorch. The full source code, together with scripts that reproduce every figure and table of this section, is available at <https://github.com/yumiharu-nakano/kernelMFG-RFU>. The Gaussian kernel $K(x, y) = e^{-\alpha|x-y|^2}$ is used throughout for the terminal MMD penalty, and the congestion kernel $W(x, y) = e^{-\alpha w|x-y|^2}$ (same family) for the running interaction cost in the EV experiments of Section 5.6. The bandwidth α is specified in each subsection; for the high-dimensional Schrödinger-bridge experiments we adopt the standard scaling $\alpha = 1/d$, while for the low-dimensional ($d = 2$) estimator-property and bimodal experiments we use $\alpha = 1$.

We organize the experiments as follows. Sections 5.1–5.5 treat the Schrödinger-bridge specialization ($c = 0$ in the potential MFG (4.2), Corollary 4.7) as a controlled warm-up of the framework that verifies the estimator and convergence theorems and demonstrates the $O(NM)$ computational scaling. Section 5.6 then presents the EV charging fleet potential MFG ($c > 0$), the main MFG application of the paper.

5.1 Estimator properties

We first verify the two key properties of the proposed RF U -statistic estimator established in Theorem 3.1: unbiasedness and the variance bound $\text{Var}(\hat{\gamma}_{M,N}^2) = O(1/M) + O(1/N)$.

5.1.1 Bias comparison

Before presenting the SBP experiments, we verify the unbiasedness of the proposed estimator. We compare three estimators of $\gamma_K(\mu, \nu)^2$: (i) the kernel U -statistic (1.3) with $O(N^2)$ complexity, (ii) the standard RFF V -statistic (Rahimi–Recht [27]) with $O(NM)$ complexity, and (iii) the proposed RF U -statistic (3.4) with $O(NM)$ complexity. For $d = 2$, $N = 200$, $M = 200$, and $\alpha = 1$, we compute each estimator 2000 times with $\mu = \nu = \mathcal{N}(0, I_2)$ so that $\gamma_K^2 = 0$.

Table 5.1: Bias comparison with $\mu = \nu = \mathcal{N}(0, I_2)$ (true $\gamma_K^2 = 0$), 2000 trials.

Estimator	Mean	Std
Kernel U -stat (1.3)	-3×10^{-5}	3.3×10^{-3}
RFF V -stat (Rahimi–Recht)	$+8.0 \times 10^{-3}$	3.3×10^{-3}
RF U -stat (proposed) (3.4)	-2×10^{-5}	3.3×10^{-3}

The kernel U -statistic and the proposed RF U -statistic both have sample means close to zero, confirming their unbiasedness. In contrast, the RFF V -statistic exhibits a positive bias of approximately $+8.0 \times 10^{-3}$, consistent with the $O(\Phi(0)/N) = O(1/200)$ prediction of Remark 3.2. At the population level this bias produces a non-vanishing penalty floor even when $\mathcal{L}(X_1) = \mu_1$ exactly, which complicates the limit-theorem analysis as $\lambda \rightarrow \infty$ (cf. Remark 3.2). Whether this distortion affects the trained drift in practice is examined separately in Section 5.1.4; the empirical answer there is that, near the SBP optimum, the biased V -statistic and the unbiased U -statistic yield indistinguishable trained drifts.

5.1.2 Variance scaling

We numerically verify the variance decomposition of Theorem 3.1(iii), namely $\text{Var}(\hat{\gamma}_{M,N}^2) = O(1/M) + O(1/N)$, where the $O(1/M)$ term arises from the random Fourier averaging and the $O(1/N)$ term is the irreducible kernel U -statistic variance identified in the decomposition (6.3).

We take $\mu = \mathcal{N}(0, I_d)$ and $\nu = \mathcal{N}(m, I_d)$ with $m = (1, 0, \dots, 0)^\top \in \mathbb{R}^{10}$ (so the alternative case $\gamma_K^2 > 0$), $d = 10$, and $\alpha = 1/d$. For each (M, N) in the grid $M \in \{50, 100, 200, 500, 1000, 2000, 5000\}$, $N \in \{50, 100, 200, 500, 1000\}$, we compute the estimator $\hat{\gamma}_{M,N}^2$ over $T = 2000$ independent trials and report the sample variance.

A non-negative least-squares fit of the model $\text{Var} = c_1/M + c_2/N$ to the 35 cells of Table 5.2 yields

$$\text{Var}(\hat{\gamma}_{M,N}^2) \approx \frac{0.0017}{M} + \frac{0.0077}{N},$$

with both coefficients positive and the model explaining the data uniformly across the grid. Two qualitative checks support the predicted scaling: (i) at the largest $M = 5000$ (so the random-feature contribution c_1/M is negligible), linear regression of $\log \text{Var}$ on $\log(1/N)$ gives slope 1.06 (theoretical: 1); (ii) at the largest $N = 1000$, the variance plateaus near $c_2/N = 7.7 \times 10^{-6}$ as M grows, matching the floor predicted by the kernel U -statistic component.

The decomposition is also visible in Table 5.2: each row stabilizes at the irreducible $O(1/N)$ floor as $M \rightarrow \infty$, in agreement with $\hat{\gamma}_{M,N}^2 \rightarrow \bar{\gamma}_K^2(D)$ from (6.3). The sample mean across all

Table 5.2: Sample variance of $\hat{\gamma}_{M,N}^2 (\times 10^{-4})$ on a grid of (M, N) . $d = 10$, $\alpha = 1/d$, $\mu = \mathcal{N}(0, I)$, $\nu = \mathcal{N}(m, I)$ with $m_1 = 1$, over 2000 trials per cell. The sample mean is constant at 0.0256 ± 0.0003 across all cells (true $\gamma_K^2 \approx 0.0256$), confirming unbiasedness.

$M \backslash N$	50	100	200	500	1000
50	2.21	1.20	0.69	0.38	0.29
100	1.88	0.89	0.48	0.24	0.19
200	1.67	0.73	0.40	0.19	0.12
500	1.48	0.67	0.34	0.14	0.08
1000	1.53	0.69	0.32	0.13	0.07
2000	1.55	0.66	0.31	0.13	0.07
5000	1.49	0.64	0.32	0.12	0.06

$7 \times 5 \times 2000 = 7 \times 10^4$ trials is 0.0256 with cell-to-cell standard deviation below 3×10^{-4} , providing additional confirmation of the unbiasedness in the alternative case.

5.1.3 Interaction estimator: bias and variance

We now verify Theorem 3.3, the analogous properties for the *interaction* estimator $\hat{\mathcal{R}}_{M,N}[\hat{\nu}^N]$ of (3.6). We take $\nu = \mathcal{N}(0, I_d)$ in dimension $d = 2$ with the Gaussian congestion kernel $W(x, y) = \exp(-\alpha|x-y|^2)$ at $\alpha = 1.0$, for which $\mathcal{R}[\nu] = \frac{1}{2}(1+4\alpha)^{-d/2}$ has the closed form 0.1. This provides an analytic target against which the empirical estimator is compared, in parallel with Sections 5.1.1–5.1.2 for the terminal MMD² estimator.

Bias check. With $N = 200$ samples and $M = 500$ random frequencies, averaged over 2,000 independent trials, the proposed unbiased estimator $\hat{\mathcal{R}}_{M,N}$ gives mean $0.10027 \pm 1.9 \times 10^{-4}$ (standard error of the mean across trials), matching the analytic value 0.1 within Monte Carlo precision (bias $+2.7 \times 10^{-4}$, within the standard error). For comparison, the corresponding V -statistic estimator

$$\hat{V}_{M,N}^{\mathcal{R}}[\hat{\nu}^N] := \frac{\Psi(0)}{2MN^2} \sum_{r=1}^M (S_c^X(\tilde{Z}_r)^2 + S_s^X(\tilde{Z}_r)^2)$$

gives mean $0.10179 \pm 1.9 \times 10^{-4}$ with bias $+1.8 \times 10^{-3}$, very close to the theoretical prediction $\frac{\Psi(0)}{2N} = \frac{1}{400} = 2.5 \times 10^{-3}$ for the Gaussian kernel. This directly verifies Theorem 3.3 (i) and the parallel of Remark 3.2 for the interaction estimator.

Variance decomposition. On a grid of 5 sample sizes $N \in \{50, 100, 200, 500, 1000\}$ and 6 random-feature counts $M \in \{20, 50, 100, 500, 10^3, 5 \cdot 10^3\}$, each cell averaged over 500 trials, we compute the empirical variance $\text{Var}(\hat{\mathcal{R}}_{M,N})$ and fit the two-parameter model $\text{Var} \approx c_1/M + c_2/N$ by non-negative least squares. The best fit is

$$\text{Var}(\hat{\mathcal{R}}_{M,N}) \approx \frac{0.0177}{M} + \frac{0.0085}{N},$$

with both coefficients positive and the model explaining $R^2 = 0.9963$ of the variation across the 30-cell grid. This confirms the variance decomposition $O(1/M) + O(1/N)$ of Theorem 3.3 (iii), in the same form (and with comparable coefficient magnitudes) as the terminal MMD² estimator of Theorem 3.1(iii) reported in Section 5.1.2. The parallel structure of the two estimators is therefore reflected at the empirical level as well as at the analytic level, justifying the use of the same (M, N) -coupling in the convergence analysis of Section 4.2 for both the terminal MMD penalty and the kernel congestion cost.

5.1.4 Effect of the penalty bias on SBP training

The microbenchmarks above quantify the bias and variance of the estimator in isolation. We now examine how the bias affects the practical Schrödinger bridge training, where the estimator is used inside an SGD loop with $\lambda \gg 1$ over thousands of iterations. We compare the proposed unbiased RF U -statistic $\hat{\gamma}_{M,N}^2$ with the standard biased RFF V -statistic $\hat{V}_{M,N}^2 := \Phi(0) |\frac{1}{N} \sum_i \phi(X_i) - \frac{1}{N} \sum_j \phi(Y_j)|^2$ of Rahimi–Recht [27]. We test two target families and several (N, λ) regimes, keeping all other hyperparameters at the values of Section 5.2–5.3. Evaluation MMD² is computed by the unbiased kernel U -statistic in all cases so that the comparison is fair.

Unimodal target (Gaussian shift, $d = 10$). We sweep $(N, \lambda^{-1}) \in \{8, 20, 80\} \times \{10^{-3}, 10^{-5}\}$ and report mean $\pm$ standard deviation over 5 seeds. Table 5.3 shows that the V -statistic penalty produces a trained drift *statistically indistinguishable* from the U -statistic penalty, with differences well within the seed standard deviation across all six configurations. The empirical V -statistic floor $\hat{V}(\mu_1, \mu_1)$ tracks the theoretical prediction $2\Phi(0)/N$ (0.025 at $N = 80$, 0.10 at $N = 20$, 0.25 at $N = 8$), confirming that the bias is present at the predicted magnitude but does not propagate into the optimization output.

Table 5.3: Effect of the penalty bias on SBP training, unimodal Gaussian shift ($d = 10$, 5 seeds). The trained drift is statistically indistinguishable between the two penalties across all (N, λ) configurations, even though the V -stat floor $\hat{V}(\mu_1, \mu_1)$ scales as $2\Phi(0)/N$ and reaches 0.25 at $N = 8$.

N	λ^{-1}	Penalty	$\mathbb{E}[X_1]_1$	$ \mathbb{E}[X_1] - m ^2$	$\hat{V}(\mu_1, \mu_1)$
80	10^{-3}	U -stat	2.72 ± 0.14	0.15 ± 0.07	$(2.0 \pm 0.3) \times 10^{-2}$
80	10^{-3}	V -stat	2.73 ± 0.14	0.14 ± 0.07	$(2.0 \pm 0.3) \times 10^{-2}$
80	10^{-5}	U -stat	2.93 ± 0.14	0.07 ± 0.02	$(2.0 \pm 0.3) \times 10^{-2}$
80	10^{-5}	V -stat	2.93 ± 0.14	0.07 ± 0.01	$(2.0 \pm 0.3) \times 10^{-2}$
20	10^{-3}	U -stat	2.75 ± 0.11	0.14 ± 0.04	$(8.5 \pm 0.6) \times 10^{-2}$
20	10^{-3}	V -stat	2.80 ± 0.13	0.12 ± 0.03	$(8.5 \pm 0.6) \times 10^{-2}$
8	10^{-3}	U -stat	2.49 ± 0.11	0.45 ± 0.11	$(2.1 \pm 0.2) \times 10^{-1}$
8	10^{-3}	V -stat	2.64 ± 0.07	0.29 ± 0.09	$(2.1 \pm 0.2) \times 10^{-1}$

Interpretation. The same comparison repeated on the bimodal target of Section 5.2 ($N = 64$, $\lambda^{-1} = 10^{-2}$, 30 seeds) is again statistically indistinguishable ($p = 0.35$ in a one-sided Fisher exact test of mode-imbalance failure rates), so the two penalties produce empirically indistinguishable

trained drifts across batch sizes $N \in [8, 80]$ and penalty parameters $\lambda^{-1} \in [10^{-5}, 10^{-2}]$. The reason is that the bias is approximately $(1/N)[2\Phi(0) - \iint K(x, x')\nu_\theta(dx)\nu_\theta(dx') - \iint K(y, y')\mu_1(dy)\mu_1(dy)]$. Near the SBP optimum $\nu_\theta \approx \mu_1$ the two self-kernel expectations balance, so the bias is approximately constant in θ and contributes only a small additional gradient. We nevertheless adopt the unbiased U -statistic in the remainder of the paper for theoretical reasons. It eliminates the population-level penalty floor of order $\Phi(0)/N$, removes the need for a bias-correction term in the convergence analysis, and yields the variance decomposition $O(1/M) + O(1/N)$ of Theorem 3.1(iii).

5.2 Bimodal target ($d = 2$)

We solve the SBP with $\mu_0 = \delta_0$ and a bimodal target $\mu_1 = \frac{1}{2}\mathcal{N}((2, 2)^\top, \frac{1}{4}I_2) + \frac{1}{2}\mathcal{N}((-2, -2)^\top, \frac{1}{4}I_2)$ in dimension $d = 2$, with $\sigma = 0.5$. The drift $u_\theta(t, x)$ is parametrized by a network with hidden dimensions (64, 32) and LayerNorm. We compare the proposed RF U -statistic ($M = 200$) with the kernel estimator (1.3), using $N = 64$, $\lambda^{-1} = 0.01$, and $\alpha = 1.0$ for 2000 epochs under the same random seed.

Both methods successfully learn the bimodal structure of μ_1 . The terminal distributions $X_1^{(\theta)}$ split into the two modes in both cases. Averaged over 5 independent runs, the RF U -statistic achieves a terminal MMD² of $(1.4 \pm 1.0) \times 10^{-2}$, comparable to the kernel estimator’s MMD² $\approx 1.1 \times 10^{-2}$, confirming that the RF approximation does not degrade accuracy in the bimodal setting. The training times are similar at $N = 64$ (10.6s vs 10.9s), as the SDE integration dominates the per-iteration cost for small batch sizes. However, the computational advantage of the RF method grows with N (see Section 5.5).

5.3 High-dimensional Gaussian shift

We test the method in higher dimensions with $\mu_0 = \delta_0$ and $\mu_1 = \mathcal{N}(m, I_d)$ where $m = (3, 0, \dots, 0)^\top \in \mathbb{R}^d$, for $d = 10, 50, 100$. The Schrödinger bridge should steer the diffusion from the origin to a Gaussian centered at m .

The drift network uses LayerNorm and hidden dimensions that scale with d . The hyperparameters are summarized in Table 5.4. The SDE is integrated via the Euler–Maruyama scheme with 20 time steps.

Table 5.4: Hyperparameters for the high-dimensional SBP experiments.

d	M	N	Epochs	Hidden dims	α	λ^{-1}
10	400	80	4000	(128, 64)	0.1	0.001
50	800	80	5000	(256, 128)	0.02	5×10^{-4}
100	1500	80	6000	(512, 256)	0.01	3×10^{-4}

The results are shown in Table 5.5. In all dimensions, the first coordinate $\mathbb{E}[X_1]_1$ approaches the target 3.0, the remaining coordinates stay near zero, and the standard deviation is close to the target value of 1.0. The MMD² decreases to the order of 10^{-3} , confirming the distributional constraint is approximately satisfied. The standard deviations across seeds are small (between 0.05 and 0.07 for $\mathbb{E}[X_1]_1$ in all three dimensions), indicating that the method is stable and the reported

Table 5.5: Results for the high-dimensional SBP (Gaussian shift, target mean $m_1 = 3.0$). Values are mean $\pm$ standard deviation over 5 independent runs with different random seeds, for each d .

d	$\mathbb{E}[X_1]_1$	$\frac{1}{d-1} \sum_{k=2}^d \mathbb{E}[X_1]_k$	Std (mean)	MMD ²
10	2.69 ± 0.07	0.002 ± 0.025	0.98 ± 0.00	$(1.4 \pm 0.2) \times 10^{-3}$
50	2.64 ± 0.07	0.002 ± 0.007	0.96 ± 0.00	$(2.1 \pm 0.2) \times 10^{-3}$
100	2.71 ± 0.05	-0.001 ± 0.002	0.95 ± 0.00	$(2.0 \pm 0.3) \times 10^{-3}$

performance is not seed-specific. The accuracy is remarkably consistent across dimensions, with $\mathbb{E}[X_1]_1$ in the range 2.64–2.71 across $d = 10, 50, 100$.

The residual gap between $\mathbb{E}[X_1]_1$ and the target 3.0 has two sources. First, with finite penalty weight λ the optimal penalised solution does not exactly match μ_1 , a gap that shrinks as $\lambda^{-1} \rightarrow 0$ (verified in Section 5.4.1). Second, the bandwidth scaling $\alpha = 1/d$ keeps the cosine argument $\text{Var}(Z^\top X | X) = 2\alpha|X|^2$ at $O(1)$ for high-dimensional targets with $\mathbb{E}|X|^2 = O(d)$, which is necessary for the RF features to remain informative and for the $O(1/M) + O(1/N)$ variance bound of Theorem 3.1(iii) to be free of hidden d -dependent constants. The price is a flatter kernel that reduces MMD discriminative power, so the same MMD² value corresponds to a larger distributional discrepancy in higher dimensions. The residual gap $\mathbb{E}[X_1]_1 \approx 2.7$ in Table 5.5 is dominated by the finite λ rather than by $\alpha = 1/d$. At $\lambda^{-1} = 10^{-4}$ the same $d = 100$ configuration reaches $\mathbb{E}[X_1]_1 = 2.91$ (Section 5.4), closing 96% of the gap. Overall, the experiments demonstrate that the proposed method enables Schrödinger bridge computation in dimensions well beyond the $d \leq 2$ range of the $O(N^2)$ kernel method in [24].

5.4 Convergence verification

The high-dimensional experiments of Section 5.3 exhibit a residual gap between $\mathbb{E}[X_1]_1$ and the target 3.0. We now diagnose this gap experimentally by sweeping the two parameters that control the approximation: the penalty weight λ (Theorem 4.6) and the number of random Fourier features M (Corollary 4.12).

5.4.1 Penalty parameter sweep

We numerically verify the convergence statement of Theorem 4.6 (i),

$$\lim_{n \rightarrow \infty} \sqrt{\lambda_n} \gamma_K(\mathcal{L}(X_1^{(n)}), \mu_1) = 0,$$

by sweeping the penalty parameter for the Gaussian shift target ($\mu_1 = \mathcal{N}(m, I_d)$ with $m = (3, 0, \dots, 0)^\top$) at $d = 10$ and $d = 50$. All other settings (network architecture, bandwidth, M , N , 20 Euler–Maruyama steps, training epochs) are kept identical to those of Section 5.3. We vary $\lambda^{-1} \in \{10^{-2}, 3 \times 10^{-3}, 10^{-3}, 3 \times 10^{-4}, 10^{-4}\}$ and report mean and standard deviation over 3 random seeds.

The results in Table 5.6 confirm the predicted convergence behaviour. As λ^{-1} decreases (i.e., λ increases) by two orders of magnitude, the terminal MMD² decreases by approximately 30-fold,

Table 5.6: λ -sweep for the $d = 10$ Gaussian shift SBP ($\mu_1 = \mathcal{N}(m, I_{10})$, $m_1 = 3.0$). Mean $\pm$ standard deviation over 3 seeds, 5000 epochs each. The control cost denotes the empirical estimate of $\mathbb{E}[\int_0^1 |u|^2 dt]$ at the end of training.

λ^{-1}	$\mathbb{E}[X_1]_1$	MMD ²	$ \mathbb{E}[X_1] - m ^2$	Control cost
10^{-2}	1.78 ± 0.05	$(2.4 \pm 0.2) \times 10^{-2}$	1.51 ± 0.12	7.4 ± 0.3
3×10^{-3}	2.28 ± 0.05	$(5.2 \pm 1.2) \times 10^{-3}$	0.54 ± 0.08	10.1 ± 0.3
10^{-3}	2.58 ± 0.06	$(2.0 \pm 0.7) \times 10^{-3}$	0.21 ± 0.05	11.6 ± 0.4
3×10^{-4}	2.76 ± 0.06	$(1.1 \pm 0.3) \times 10^{-3}$	0.087 ± 0.028	13.0 ± 0.7
10^{-4}	2.81 ± 0.06	$(0.8 \pm 0.2) \times 10^{-3}$	0.061 ± 0.020	13.8 ± 0.9

the squared mean error $|\mathbb{E}[X_1] - m|^2$ decreases by approximately 25-fold, and the first-coordinate mean $\mathbb{E}[X_1]_1$ approaches the target value 3.0 from below. Computing $\sqrt{\lambda} \gamma_K \leq \sqrt{\lambda} \cdot \text{MMD}^2$ from the table gives 1.51 at $\lambda^{-1} = 10^{-2}$ and 0.087 at $\lambda^{-1} = 10^{-4}$, a 17-fold reduction consistent with $\sqrt{\lambda} \gamma_K \rightarrow 0$ as stated in Theorem 4.6 (i). The control cost simultaneously increases monotonically from 7.4 to 13.8, exhibiting the expected fidelity–cost trade-off: stronger enforcement of the terminal constraint requires a more aggressive control.

The residual gap of $\mathbb{E}[X_1]_1 \approx 2.81$ to the target 3.0 at $\lambda^{-1} = 10^{-4}$ is no longer dominant, and could in principle be eliminated by further decreasing λ^{-1} at the cost of additional iterations. This experiment thus identifies the finite penalty weight as the principal source of the residual gap discussed in Section 5.3.

To confirm that the same diagnosis applies in higher dimensions and that the residual gap $\mathbb{E}[X_1]_1 \approx 2.7$ observed in Table 5.5 for $d \geq 10$ is not a structural limitation of the $\alpha = 1/d$ scaling, we repeat the sweep at $d = 100$ with the corresponding hyperparameters of Table 5.4 ($\alpha = 0.01$, $M = 1500$, $N = 80$, 6000 epochs); the analogous sweep at the intermediate $d = 50$ is qualitatively identical and is omitted to save space.

Table 5.7: λ -sweep for the $d = 100$ Gaussian shift SBP ($\mu_1 = \mathcal{N}(m, I_{100})$, $m_1 = 3.0$). Mean $\pm$ standard deviation over 3 seeds, 6000 epochs each, $\alpha = 0.01$, $M = 1500$.

λ^{-1}	$\mathbb{E}[X_1]_1$	MMD ²	$ \mathbb{E}[X_1] - m ^2$	Control cost
10^{-2}	0.96 ± 0.02	$(1.1 \pm 0.0) \times 10^{-1}$	4.223 ± 0.099	4.4 ± 0.5
3×10^{-3}	1.72 ± 0.03	$(3.6 \pm 0.1) \times 10^{-2}$	1.774 ± 0.089	16.6 ± 1.4
10^{-3}	2.28 ± 0.01	$(9.0 \pm 0.4) \times 10^{-3}$	0.740 ± 0.014	30.8 ± 2.0
3×10^{-4}	2.72 ± 0.06	$(2.0 \pm 0.1) \times 10^{-3}$	0.321 ± 0.029	43.3 ± 3.6
10^{-4}	2.91 ± 0.01	$(1.3 \pm 0.1) \times 10^{-3}$	0.263 ± 0.034	51.8 ± 4.5

Table 5.7 shows that the trend in $d = 100$ is identical to and, if anything, more favourable than in $d = 10$. $\mathbb{E}[X_1]_1$ progresses from 0.96 ± 0.02 at $\lambda^{-1} = 10^{-2}$ to 2.91 ± 0.01 at $\lambda^{-1} = 10^{-4}$, closing approximately 96% of the gap to the target 3.0. The best value of $\mathbb{E}[X_1]_1$ at $\lambda^{-1} = 10^{-4}$ thus lies in the narrow range 2.81 (at $d = 10$) to 2.91 (at $d = 100$), demonstrating that the residual gap of $\mathbb{E}[X_1]_1 \approx 2.7$ reported at the working λ^{-1} values of Table 5.5 is driven by the choice of λ rather

than by a degradation of the method with dimension. The price of pushing λ further is a roughly proportional increase in the control cost (at $d = 100$: from 4.4 to 51.8, a 12-fold increase) and in the number of optimization iterations needed for convergence, consistent with the trade-off quantified in the explicit a.s. rate (4.11) of Proposition 4.9.

5.4.2 Kernel U -statistic limit ($M \rightarrow \infty$)

We numerically verify Corollary 4.12, which states that the random Fourier U -statistic penalty converges to the kernel U -statistic penalty $\bar{\gamma}_K^2(D)$ as $M \rightarrow \infty$, and that both yield the same limiting Schrödinger bridge. Using the $d = 10$ Gaussian shift setting of Section 5.3 as a controlled benchmark ($N = 80$, $\lambda^{-1} = 10^{-3}$, 5000 epochs, 5 seeds, all other hyperparameters unchanged), we compare four penalty estimators: the kernel U -statistic $\bar{\gamma}_K^2(D)$ of complexity $O(N^2)$, and the RF U -statistic $\hat{\gamma}_{M,N}^2$ for $M \in \{100, 400, 1600\}$. Evaluation is performed by the kernel U -statistic in all cases for fair comparison.

Table 5.8: Kernel U -statistic vs RF U -statistic penalty in SBP training ($d = 10$ Gaussian shift, $N = 80$, $\lambda^{-1} = 10^{-3}$, 5000 epochs, 5 seeds). Evaluation MMD² is computed by the kernel U -statistic for fair comparison.

Method	$\mathbb{E}[X_1]_1$	$ \mathbb{E}[X_1] - m ^2$	MMD ² (eval)	Std (mean)
Kernel U -stat	2.69 ± 0.08	0.13 ± 0.04	$(1.40 \pm 0.70) \times 10^{-3}$	0.98 ± 0.01
RF U -stat ($M = 100$)	2.63 ± 0.09	0.18 ± 0.07	$(2.11 \pm 0.88) \times 10^{-3}$	0.98 ± 0.01
RF U -stat ($M = 400$)	2.67 ± 0.12	0.16 ± 0.07	$(1.98 \pm 0.77) \times 10^{-3}$	0.98 ± 0.01
RF U -stat ($M = 1600$)	2.77 ± 0.09	0.10 ± 0.05	$(2.07 \pm 0.84) \times 10^{-3}$	0.97 ± 0.00

The four methods produce statistically indistinguishable solutions. The first-coordinate means $\mathbb{E}[X_1]_1$ all fall in $[2.63, 2.77]$, well within the seed-to-seed standard deviation, and the standard deviations across the remaining coordinates agree to two decimal places. The MMD² values for the RF estimators ($M = 100, 400, 1600$) cluster around 2×10^{-3} , slightly above the kernel U -statistic's 1.4×10^{-3} , but again within seed variability. This empirical equivalence supports Corollary 4.12: the RF approximation is a faithful surrogate of the kernel U -statistic in the SBP training loop, and increasing M does not push the RF method beyond the kernel U -statistic baseline.

A complementary cost analysis is given in Section 5.5, where we show that for sufficiently large batch size N the kernel U -statistic computation ($O(N^2)$ per iteration, with backpropagation through the full $N \times N$ kernel matrix) becomes a memory and compute bottleneck, whereas the RF U -statistic remains scalable at $O(NM)$ cost. For the moderate $N = 80$ used here, both estimators have comparable per-iteration cost, so the table demonstrates only the equivalence in solution quality. The computational advantage of the RF method emerges in higher dimensions and larger batch sizes.

5.5 Computational scaling

We compare the proposed RF U -statistic with the kernel estimator (1.3) in terms of both solution quality and computational cost.

We measure the per-iteration wall-clock time of the MMD computation alone (forward + backward, $d = 10$, $\alpha = 0.1$, $M = 500$), which isolates the kernel-evaluation cost from the SDE overhead.

Table 5.9: MMD computation time (forward + backward, in milliseconds; $d = 10$, $\alpha = 0.1$, $M = 500$). Solution quality of the two estimators is comparable in the SBP training pipeline (verified in Section 5.4.2 for $d = 10$).

N	Kernel $O(N^2)$	RF U -stat $O(NM)$	Speedup
100	0.6	0.7	$0.8\times$
200	0.8	0.6	$1.3\times$
500	4.7	1.1	$4.1\times$
1000	19.3	2.0	$9.5\times$
2000	83.3	3.7	$22.3\times$

The speedup reaches $22\times$ at $N = 2000$, confirming the $O(N^2)$ vs $O(NM)$ scaling. The kernel method also requires storing the $N \times N$ kernel matrix in the computation graph for backpropagation through the SDE trajectory, which leads to memory exhaustion for large N on typical hardware; the RF U -statistic avoids this entirely, with memory footprint $O(NM + Nd)$. This memory advantage is essential for high-dimensional MFGs and SBPs where both large N (for accurate sampling) and large d are required simultaneously.

5.6 Potential MFG: EV charging fleet coordination

We now turn to the main MFG application of the paper. Consider a fleet of electric vehicles that need to charge over an operational horizon $[0, T]$ with $T = 1$. Each EV is described by a state $X_t = (s_t, h_t) \in \mathbb{R} \times \mathbb{R}$ where $s_t \in \mathbb{R}$ is the state of charge (SOC), i.e., the ratio of current stored energy to the nominal battery capacity, normalized so that $s_t = 0$ corresponds to a fully discharged battery and $s_t = 1$ to a fully charged one. Further, $h_t \in \mathbb{R}$ is a *physical heterogeneity coordinate* representing the log of the per-vehicle charging-speed multiplier, i.e., $\eta(h) := e^h$ scales the actual SOC change rate produced by a unit demand, modelling the combined effect of battery capacity, charger maximum power, and conversion efficiency. The controlled dynamics are

$$(5.1) \quad ds_t = \eta(h) u_\theta(t, X_t) dt + \sigma dB_t, \quad dh_t = 0, \quad s_0 \sim \mathcal{N}(0.20, 0.05^2), \quad h_0 \sim \mathcal{N}(0, \sigma_h^2),$$

with $\sigma = 0.05$ and $\sigma_h = 0.3$, so that $\eta(h)$ is log-normal with mean 1 and approximate 95% range $[0.55, 1.8]$. The control $u_\theta(t, X_t) \in \mathbb{R}$ is the EV's demand fraction (read by the on-board charger and converted into an actual SOC rate $\eta(h)u_\theta$); it is parametrised by a neural network that takes (t, s, h) as input, so the drift can differentiate charging schedules across EVs with different physical $\eta(h)$.

Following classical formulations of mean-field-type EV charging coordination [11, 21, 26], we adopt an *aggregate-demand-price* congestion cost. Let $u^* : [0, 1] \rightarrow [0, 1]$ be the SOC-dependent charging-power profile of a Li-ion battery: u^* is approximately maximal in the constant-current range (20–80% SOC, drawing tens of kW per vehicle) and tapered in the constant-voltage range ($\gtrsim 80\%$ SOC). We use the smooth profile

$$(5.2) \quad u^*(s) = \sigma(\beta(s - s_-)) \sigma(\beta(s_+ - s)), \quad \beta = 20, \quad s_- = 0.1, \quad s_+ = 0.85,$$

with $\sigma(x) := 1/(1 + e^{-x})$, which is bounded continuous and captures the CC plateau and CV taper ($u^*(s) \approx 1$ for $s \in [0.2, 0.8]$, $u^*(0.85) \approx 0.5$, $u^*(0.95) \approx 0.18$, $u^*(s) \rightarrow 0$ as $s \rightarrow 0$ or $s \rightarrow 1$). The grid power drawn by vehicle i is then $\eta(h^{(i)})u^*(s_t^{(i)})$ (charging-speed multiplier times the SOC-dependent profile), and the cost functional is the potential MFG (4.2) with

- Terminal target $\mu_T = \mathcal{N}(0.85, 0.05^2)$ on the SOC component (deadline distribution).
- Terminal kernel $K(s, s') = e^{-\alpha_K(s-s')^2}$ with $\alpha_K = 50$ on the SOC coordinate only.
- Aggregate-demand congestion

$$(5.3) \quad \mathcal{R}^{(D)}[\nu_t] = D[\nu_t]^2, \quad D[\nu] := \int \eta(h) u^*(s) \nu(ds, dh),$$

representing the squared aggregate charging demand of the fleet at time t .

- Weights $\lambda = 10^3$ and $c \in \{0, 10, 100\}$ (no-congestion baseline, mild congestion, strong congestion).

The cost $\mathcal{R}^{(D)}[\nu_t]$ corresponds to a quadratic price function applied to the aggregate fleet demand, and equals the interaction integral $\iint \eta(h)u^*(s) \eta(h')u^*(s') \nu_t(ds, dh) \nu_t(ds', dh')$ with the rank-one kernel $(x, x') \mapsto \eta(h)u^*(s) \cdot \eta(h')u^*(s')$. Two EVs both in the constant-current phase with similar η contribute jointly to grid demand and produce a synchronized peak load, whereas an EV in the CV phase ($u^* \rightarrow 0$) or with very small η contributes little and is not aggregated.

Empirical estimator and convergence theory. $\mathcal{R}^{(D)}[\nu]$ falls outside the kernel-MMD framework of Theorem 3.3 (the rank-one kernel $\eta(h)u^*(s) \cdot \eta(h')u^*(s')$ is not translation invariant), but the functional $\nu \mapsto D[\nu]^2$ is non-negative and continuous in the weak topology on $\mathcal{P}(\mathbb{R}^2)$, since $(s, h) \mapsto \eta(h)u^*(s)$ is bounded continuous (recall η is bounded above on the support of h_0 for any finite training horizon, and the SDE preserves h). Writing $g_i := \eta(h^{(i)})u^*(s_t^{(i)})$, an unbiased U -statistic estimator with $O(N)$ complexity is available:

$$(5.4) \quad \hat{\mathcal{R}}_N^{(D)}[\hat{\nu}_t^N] := \frac{1}{N(N-1)} \sum_{i \neq j} g_i g_j = \frac{(\sum_i g_i)^2 - \sum_i g_i^2}{N(N-1)},$$

which is strongly consistent ($\hat{\mathcal{R}}_N^{(D)}[\hat{\nu}_t^N] \rightarrow \mathcal{R}^{(D)}[\nu]$ a.s. as $N \rightarrow \infty$) by the standard SLLN for U -statistics [30, Section 5.4]. The sample-level almost-sure convergence of the empirical MFG to the constrained value therefore follows from Proposition 4.8 (no rate condition on a running-cost feature count is required, only $N_n \rightarrow \infty$).

State-space idealization. The SOC is physically constrained to $[0, 1]$, but the chosen means and variances keep $\mathbb{P}(s_t \notin [0, 1])$ below 1.4×10^{-3} ; we therefore treat the SOC as a real-valued diffusion, leaving a $[0, 1]$ -truncated and reflecting variant for future work.

Training uses $N = 128$ paths per batch, $M = 300$ random frequencies for the terminal kernel K (the aggregate-demand cost (5.4) requires no random features), 20 Euler–Maruyama steps, learning rate 10^{-3} , and 2000 epochs. We report mean $\pm$ standard deviation over 3 random seeds.

Table 5.10: EV charging fleet potential MFG with physical heterogeneity ($\eta(h) = e^h$, $\sigma_h = 0.3$) and aggregate-demand congestion, $\lambda = 10^3$, 3 seeds per configuration. “Peak D ” is $\max_{t \in (0, T]} \hat{D}[\hat{\nu}_t^N]$ on a held-out batch of $n_{\text{eval}} = 2000$ paths, “Mean D ” is the time-averaged $\hat{D}[\hat{\nu}_t^N]$ on the same paths, and the evaluation MMD² on SOC uses the kernel U -statistic for fairness. “ $\|u_n\|_\infty$ ” is the empirical L^∞ norm of the trained drift on the evaluation batch, verifying the uniform bound used in Proposition 4.11. $c = 0$ is the no-congestion baseline; $c = 10, 100$ activate the aggregate-demand congestion cost (5.3).

c	$\mathbb{E}[s_T]$	SOC std at T	Eval. MMD ²	Peak D	Mean D	$\ u_n\ _\infty$
0	0.844 ± 0.003	0.055	$(1.6 \pm 1.7) \times 10^{-3}$	1.044 ± 0.001	0.946	1.10 ± 0.02
10	0.844 ± 0.002	0.055	$(1.6 \pm 1.1) \times 10^{-3}$	1.044 ± 0.001	0.939	1.10 ± 0.02
100	0.813 ± 0.035	0.236	$(3.8 \pm 3.3) \times 10^{-3}$	0.887 $\pm$ 0.168	0.431	2.48 ± 0.20

Table 5.10 shows three effects. *First*, the framework solves the EV charging MFG: $\mathbb{E}[s_T] \approx 0.844$ at $c = 0, 10$, with a small downward bias to 0.81 at $c = 100$ reflecting the congestion-vs-terminal trade-off. *Second*, the aggregate-demand congestion cost reduces both peak and time-averaged demand: at $c = 100$, peak D drops from 1.044 to 0.887 ($\approx 15\%$ peak shaving) and mean D from 0.946 to **0.431** ($\approx 54\%$ reduction), at the price of a moderate increase in evaluation MMD² ($1.6 \times 10^{-3} \rightarrow 3.8 \times 10^{-3}$, factor ≈ 2.4). At $c = 10$ the congestion weight is too small relative to $\lambda = 10^3$ to differ from the $c = 0$ baseline; the effect emerges only at $c = 100$. *Third*, this reduction combines two ingredients: the physical heterogeneity $\eta(h) = e^h$ naturally desynchronises fast ($\eta > 1$) and slow ($\eta < 1$) chargers, and the drift $u_\theta(t, s, h)$ further staggers them in time. The combined effect broadens the terminal SOC distribution at $c = 100$ ($\text{std}(s_T) : 0.055 \rightarrow \mathbf{0.236}$, factor 4.3), showing that the strong-congestion MFG sacrifices terminal SOC concentration to flatten grid demand.

Empirical verification of the uniform drift bound. The rate analysis of Proposition 4.11 assumes a uniform L^∞ bound on the trained drift sequence $\{u_n\}$. The rightmost column of Table 5.10 reports $\|u_n\|_\infty := \max_t \|u_n(t, \cdot)\|_{L^\infty}$ measured on the evaluation batch. The largest value occurs at the strong-congestion setting $c = 100$, where the drift must work hardest to spread the population in time. The uniform bound is satisfied without any explicit weight clipping or bounded-output activation in the present network architecture (hidden dimensions (64, 32) with ReLU activations and LayerNorm).

Scope and physical realism. The present demonstration adopts a single physical heterogeneity coordinate h encoding the log charging-speed multiplier; extending to richer physical heterogeneities (battery capacity, plug-out time uncertainty, comfort preference) that act jointly through h -dependent dynamics and a domain-justified law on h is a natural problem-driven direction. The unbiased U -statistic estimator (5.4) keeps the per-iteration cost of the congestion evaluation at $O(N)$ and dimension-free in the heterogeneity dimension, so such extensions are supported by the framework without algorithmic modification.

5.7 Positioning against deep-learning MFG / SBP baselines

The methodology developed in this paper is one of several deep-learning approaches to high-dimensional MFG and SBP that have appeared in recent years. The most directly relevant baselines are APAC-Net of Lin et al. [20], which alternates between a population network and a control network in a saddle-point formulation; the HJB-based machine-learning framework of Ruthotto et al. [28], which parametrises the value function rather than the control; the deep MFG/MFC methods surveyed in [8, 9]; and the bridge-matching and diffusion-Schrödinger-bridge methods [12, 35, 31, 34], which target the $c = 0$ specialization. Table 5.11 summarises the structural differences relevant to a JSC-style assessment.

Table 5.11: Structural comparison with representative deep-learning MFG and SBP methods. “Networks” counts the neural networks involved in the optimization; “Cost class” indicates the running- and terminal-cost structure each method supports; “Convergence” refers to the strongest known theoretical guarantee for the method as actually implemented (rather than for an idealized population-level limit); “Per-iter” is the asymptotic per-iteration cost scaling (M = random-feature count, N = batch size, τ = SDE time steps, P_θ = network parameter count).

Method	Reference	Networks	Cost class	Convergence	Per-iter
Ours	this paper	1 (drift)	MMD + weakly cts. $\mathcal{R}$	sample-level a.s. + rate	$O(NM + \tau P_\theta)$
APAC-Net	[20]	2 (pop. + ctrl.)	Lagrangian, no soft target	PDE-level (asymptotic)	$O(\tau P_\theta)$
ML-MFG	[28]	1 (value fn.)	local + interaction (HJB form)	PDE-level (asymptotic)	$O(\tau P_\theta)$
Deep MFG/MFC	[8, 9]	2-3	general $f(\cdot, \nu)$, $g(\cdot, \nu)$	population-level	$O(\tau P_\theta)$
DSB / DSBM	[12, 31]	2 (score)	SBP only ($c = 0$)	empirical	$O(\tau P_\theta)$
SB max-lik.	[35]	2	SBP only ($c = 0$)	empirical	$O(\tau P_\theta)$

Three structural distinctions are worth highlighting. *First*, our method uses a single drift network, whereas APAC-Net, DSB, DSBM and the deep MFG/MFC variants use two or more (e.g., a control network and a population/score network, or a value function and a forward-backward pair). The reason is structural. The terminal MMD penalty $\gamma_K^2(\mathcal{L}(X_T^{(n)}), \mu_T)$ is a functional of the controlled marginal $\mathcal{L}(X_T^{(n)})$ that can be estimated directly from sampled controlled paths, without learning the density or score of $\mathcal{L}(X_T^{(n)})$. In contrast, deep-learning MFG/SBP methods that target a hard distributional constraint or a score-matching objective need an explicit estimate of the population density or score, which is provided by an auxiliary population or score network. *Second*, our convergence theorem operates at the *sample* level for the empirical objective that is actually minimized (Theorem 4.6 and Proposition 4.11), whereas the comparison methods either provide population-level guarantees (deep MFG/MFC of [8]) or no explicit convergence statement in the sampled regime (APAC-Net, DSB, DSBM). *Third*, our per-iteration cost on the kernel-MMD terms is the explicit $O(NM)$ of the random-feature representation; the comparison methods rely on neural-network forward/backward passes whose cost is implicit in P_θ .

Direct numerical comparison. A fair head-to-head benchmark with the above baselines is non-trivial because of differing problem formulations: APAC-Net and Ruthotto et al. solve HJB-style problems without a soft distributional terminal target, and DSB / DSBM target the SBP specialisation $c = 0$ via score-matching rather than a penalty. The deep MFG/MFC framework of

[8, 9] admits per-agent costs $f(t, x, \alpha, \nu)$, $g(x, \nu)$ and in fact subsumes our kernel-MMD terminal cost as a special $g(x, \nu)$; adopting that methodology would, however, replace our unbiased $O(NM)$ estimator by a biased sample mean, introduce a separate population or value-function network, and yield population-level rather than sample-level guarantees. The generalised conditional-gradient method of [23, 22] proves non-asymptotic convergence rates for potential MFGs with local coupling on fully discretised finite-difference grids, which is complementary in scope: their analysis is sharp on the grid but inherits the curse of dimensionality, whereas our sample-level rate (Proposition 4.11) is mesh-free and applies to nonlocal kernel-MMD and aggregate-demand couplings. A quantitative benchmark on a common instance well suited to all candidate methods is left for future work; Section 5.5 gives the within-framework comparison of the RF U -statistic against the $O(N^2)$ kernel method of [24].

6 Proofs

6.1 Proof of Theorem 3.1

We prove each part separately.

(i) Unbiasedness. Fix $z \in \mathbb{R}^d$. By expanding the squares in (3.2),

$$\begin{aligned} (S_c^X(z))^2 + (S_s^X(z))^2 &= \sum_{i=1}^N \sum_{j=1}^N (\cos(z^\top X_i) \cos(z^\top X_j) + \sin(z^\top X_i) \sin(z^\top X_j)) \\ &= \sum_{i=1}^N \sum_{j=1}^N \cos(z^\top (X_i - X_j)) = N + \sum_{\substack{i,j=1 \\ i \neq j}}^N \cos(z^\top (X_i - X_j)), \end{aligned}$$

where we used $\cos(a - b) = \cos a \cos b + \sin a \sin b$ and $\cos(0) = 1$. Since $X_1, \dots, X_N$ are i.i.d. from μ and X, X' denote independent copies with law μ ,

$$\sum_{\substack{i,j=1 \\ i \neq j}}^N \mathbb{E} [\cos(z^\top (X_i - X_j))] = N(N-1) \mathbb{E} [\cos(z^\top (X - X'))].$$

Therefore

$$(6.1) \quad \mathbb{E}[U_{XX}(z)] = \frac{\mathbb{E}[(S_c^X(z))^2 + (S_s^X(z))^2 - N]}{N(N-1)} = \mathbb{E}[\cos(z^\top (X - X'))].$$

Similarly, $\mathbb{E}[U_{YY}(z)] = \mathbb{E}[\cos(z^\top (Y - Y'))]$ where Y, Y' are independent copies with law ν .

For the cross term (3.3), since (X_i, Y_j) are independent for all pairs,

$$\mathbb{E}[V_{XY}(z)] = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N \mathbb{E} [\cos(z^\top (X_i - Y_j))] = \mathbb{E} [\cos(z^\top (X - Y))].$$

Therefore, for each fixed z ,

$$\mathbb{E}[U_{XX}(z) - 2V_{XY}(z) + U_{YY}(z)] = \mathbb{E}[\cos(z^\top (X - X'))] - 2\mathbb{E}[\cos(z^\top (X - Y))] + \mathbb{E}[\cos(z^\top (Y - Y'))].$$

Now, averaging over $Z_1, \dots, Z_M$ and applying the tower property, we have

$$\begin{aligned}\mathbb{E}[\hat{\gamma}_{M,N}^2] &= \frac{\Phi(0)}{M} \sum_{r=1}^M \mathbb{E}[\mathbb{E}[U_{XX}(Z_r) - 2V_{XY}(Z_r) + U_{YY}(Z_r) \mid Z_r]] \\ &= \Phi(0) \mathbb{E}[\mathbb{E}[\cos(Z^\top(X - X')) \mid Z] - 2\mathbb{E}[\cos(Z^\top(X - Y)) \mid Z] + \mathbb{E}[\cos(Z^\top(Y - Y')) \mid Z]] \\ &= \gamma_K(\mu, \nu)^2,\end{aligned}$$

where the last step uses the representation (2.6).

(ii) Complexity. For each frequency $z = Z_r$, computing $S_c^X(z)$ and $S_s^X(z)$ requires $O(N)$ operations, namely evaluating $\cos(z^\top X_i)$ and $\sin(z^\top X_i)$ for $i = 1, \dots, N$ and summing. The same complexity holds for $S_c^Y(z)$ and $S_s^Y(z)$. From these four sums, $U_{XX}(z)$, $V_{XY}(z)$, and $U_{YY}(z)$ are each computed in $O(1)$. Since there are M frequencies, the total cost is $O(NM)$.

(iii) Variance. Let

$$g(z) := \Phi(0)[U_{XX}(z) - 2V_{XY}(z) + U_{YY}(z)],$$

so that $\hat{\gamma}_{M,N}^2 = \frac{1}{M} \sum_{r=1}^M g(Z_r)$. Note that $g(z)$ depends on the data $D := (X_1, \dots, X_N, Y_1, \dots, Y_N)$, which is shared across all frequencies $Z_1, \dots, Z_M$. Hence $g(Z_1), \dots, g(Z_M)$ are conditionally independent given D . Conditioning on D via the law of total variance, we get

$$(6.2) \quad \text{Var}(\hat{\gamma}_{M,N}^2) = \mathbb{E}[\text{Var}(\hat{\gamma}_{M,N}^2 \mid D)] + \text{Var}(\mathbb{E}[\hat{\gamma}_{M,N}^2 \mid D]).$$

Let us observe the conditional term. Given D , the frequencies $Z_1, \dots, Z_M$ are i.i.d., so

$$\text{Var}(\hat{\gamma}_{M,N}^2 \mid D) = \frac{1}{M} \text{Var}(g(Z_1) \mid D).$$

From (3.2)–(3.3) and the trivial bounds $|U_{XX}(z)|, |V_{XY}(z)|, |U_{YY}(z)| \leq 1$, $|g(z)| \leq 4\Phi(0)$ uniformly in z and D . Hence $\text{Var}(g(Z_1) \mid D) \leq 16\Phi(0)^2$, and therefore $\mathbb{E}[\text{Var}(\hat{\gamma}_{M,N}^2 \mid D)] = O(1/M)$.

As for the marginal term, a direct computation of the conditional expectation $\mathbb{E}[g(Z_1) \mid D]$ gives

$$\begin{aligned}(6.3) \quad \mathbb{E}[g(Z_1) \mid D] &= \Phi(0)\mathbb{E}[U_{XX}(Z) - 2V_{XY}(Z) + U_{YY}(Z) \mid D] \\ &= \frac{1}{N(N-1)} \sum_{\substack{i,j=1 \\ i \neq j}}^N K(X_i, X_j) - \frac{2}{N^2} \sum_{i,j=1}^N K(X_i, Y_j) + \frac{1}{N(N-1)} \sum_{\substack{i,j=1 \\ i \neq j}}^N K(Y_i, Y_j) \\ &=: \bar{\gamma}_K^2(D),\end{aligned}$$

where we used $\Phi(0)\mathbb{E}[\cos(Z^\top(x - y))] = \Phi(x - y) = K(x, y)$ by (2.3). Thus $\mathbb{E}[\hat{\gamma}_{M,N}^2 \mid D] = \bar{\gamma}_K^2(D)$ coincides with the kernel U -statistic estimator (1.3). It remains to bound $\text{Var}(\bar{\gamma}_K^2(D))$. Write

$$\bar{\gamma}_K^2(D) = T_{XX} - 2T_{XY} + T_{YY},$$

where

$$\begin{aligned}T_{XX} &= \frac{1}{N(N-1)} \sum_{i \neq j} K(X_i, X_j), \quad T_{YY} = \frac{1}{N(N-1)} \sum_{i \neq j} K(Y_i, Y_j), \\ T_{XY} &= \frac{1}{N^2} \sum_{i,j=1}^N K(X_i, Y_j).\end{aligned}$$

The terms T_{XX} and T_{YY} are one-sample U -statistics of degree 2 with symmetric kernel $K(\cdot, \cdot)$, and T_{XY} is a two-sample U -statistic of degree $(1, 1)$ with kernel $K(\cdot, \cdot)$.

By Hoeffding's variance formula for one-sample U -statistics [30, Section 5.2],

$$\text{Var}(T_{XX}) = \frac{2}{N(N-1)} [2(N-2)\zeta_1^X + \zeta_2^X],$$

where the Hoeffding variance components are

$$\zeta_1^X = \text{Var}(\mathbb{E}[K(X_1, X_2) | X_1]), \quad \zeta_2^X = \text{Var}(K(X_1, X_2)).$$

The uniform bound $|K(x, y)| \leq \Phi(0)$ yields $\zeta_1^X, \zeta_2^X \leq \Phi(0)^2$, hence $\text{Var}(T_{XX}) \leq 4\Phi(0)^2/N$ for $N \geq 2$. The same bound applies to T_{YY} with corresponding components $\zeta_1^Y, \zeta_2^Y \leq \Phi(0)^2$.

To estimate the cross term T_{XY} , we use the representation

$$\text{Var}(T_{XY}) = \text{Var}(\mathbb{E}[T_{XY} | X]) + \mathbb{E}[\text{Var}(T_{XY} | X)].$$

Setting $\bar{K}_\nu(x) := \int_{\mathbb{R}^d} K(x, y) \nu(dy)$, we have $\mathbb{E}[T_{XY} | X] = \frac{1}{N} \sum_{i=1}^N \bar{K}_\nu(X_i)$, so that

$$\text{Var}(\mathbb{E}[T_{XY} | X]) = \frac{1}{N} \text{Var}(\bar{K}_\nu(X_1)) \leq \frac{\Phi(0)^2}{N},$$

where we used $|\bar{K}_\nu(x)| \leq \Phi(0)$. As for the conditional variance term, given X the random variables $Y_1, \dots, Y_N$ are i.i.d., so we have

$$\text{Var}(T_{XY} | X) = \frac{1}{N^2} \sum_{j=1}^N \text{Var} \left(\frac{1}{N} \sum_{i=1}^N K(X_i, Y_j) \middle| X \right) \leq \frac{\Phi(0)^2}{N},$$

since the quantity inside the variance is bounded by $\Phi(0)$ in absolute value. Therefore, $\text{Var}(T_{XY}) \leq 2\Phi(0)^2/N$.

Combining these three estimates via the triangle inequality for the L^2 -norm, we see

$$\sqrt{\text{Var}(\hat{\gamma}_K^2(D))} \leq \sqrt{\text{Var}(T_{XX})} + 2\sqrt{\text{Var}(T_{XY})} + \sqrt{\text{Var}(T_{YY})} \leq \frac{C\Phi(0)}{\sqrt{N}}$$

for some constant $C > 0$, whence $\text{Var}(\hat{\gamma}_K^2(D)) = O(1/N)$. Consequently, substituting into (6.2), we obtain

$$\text{Var}(\hat{\gamma}_{M,N}^2) = O(1/M) + O(1/N),$$

completing the proof of the theorem.

Remark 6.1. The decompositions (6.2) and (6.3) shows that the proposed estimator is exactly the kernel U -statistic $\hat{\gamma}_K^2(D)$ averaged over random frequencies: the random Fourier representation contributes only the conditional variance term $O(1/M)$, while the irreducible sample variance $O(1/N)$ inherent to the kernel U -statistic is preserved. This decomposition is also consistent with Lemma 4.4: both the kernel U -statistic $\hat{\gamma}_K^2(D) \rightarrow \gamma_K^2$ as $N \rightarrow \infty$, and the random-feature average $\hat{\gamma}_{M,N}^2 - \hat{\gamma}_K^2(D) \rightarrow 0$, $M \rightarrow \infty$, contribute to the joint a.s. convergence used in Theorem 4.6.

6.2 Proof of Theorem 3.3

The proof follows the same template as that of Theorem 3.1 given in Section 6.1. Since the interaction $\mathcal{R}[\nu]$ involves only the X -sample, the cross terms V_{XY} and U_{YX} are absent, and (K, Φ, ρ) is replaced by (W, Ψ, ρ_W) . We prove each part separately.

(i) Unbiasedness. Fix $\tilde{z} \in \mathbb{R}^d$. By the algebraic identity used in the derivation of (3.2),

$$(S_c^X(\tilde{z}))^2 + (S_s^X(\tilde{z}))^2 = N + \sum_{\substack{i,j=1 \\ i \neq j}}^N \cos(\tilde{z}^\top (X_i - X_j)).$$

Since $X_1, \dots, X_N$ are i.i.d. from ν and X, X' denote independent copies with law ν ,

$$\sum_{\substack{i,j=1 \\ i \neq j}}^N \mathbb{E}[\cos(\tilde{z}^\top (X_i - X_j))] = N(N-1) \mathbb{E}[\cos(\tilde{z}^\top (X - X'))],$$

and therefore

$$(6.4) \quad \mathbb{E}[U_{XX}(\tilde{z})] = \frac{\mathbb{E}[(S_c^X(\tilde{z}))^2 + (S_s^X(\tilde{z}))^2 - N]}{N(N-1)} = \mathbb{E}[\cos(\tilde{z}^\top (X - X'))].$$

Averaging over $\tilde{Z}_1, \dots, \tilde{Z}_M \sim \rho_W/\Psi(0)$ and applying the tower property,

$$\begin{aligned} \mathbb{E}[\hat{\mathcal{R}}_{M,N}[\hat{\nu}^N]] &= \frac{\Psi(0)}{2M} \sum_{r=1}^M \mathbb{E}[\mathbb{E}[U_{XX}(\tilde{Z}_r) | \tilde{Z}_r]] \\ &= \frac{\Psi(0)}{2} \mathbb{E}[\mathbb{E}[\cos(\tilde{Z}^\top (X - X')) | \tilde{Z}]] = \mathcal{R}[\nu], \end{aligned}$$

where the last equality uses the Fourier representation (3.5).

(ii) Complexity. For each frequency $\tilde{z} = \tilde{Z}_r$, computing $S_c^X(\tilde{z})$ and $S_s^X(\tilde{z})$ requires $O(N)$ operations, namely evaluating $\cos(\tilde{z}^\top X_i)$ and $\sin(\tilde{z}^\top X_i)$ for $i = 1, \dots, N$ and summing. From these two sums, $U_{XX}(\tilde{z})$ is computed in $O(1)$. Since there are M frequencies and assembling (3.6) adds an $O(M)$ overhead, the total cost is $O(NM)$.

(iii) Variance. Let

$$g_W(\tilde{z}) := \frac{\Psi(0)}{2} U_{XX}(\tilde{z}),$$

so that $\hat{\mathcal{R}}_{M,N}[\hat{\nu}^N] = \frac{1}{M} \sum_{r=1}^M g_W(\tilde{Z}_r)$. Note that $g_W(\tilde{z})$ depends on the data $D_X := (X_1, \dots, X_N)$, which is shared across all frequencies $\tilde{Z}_1, \dots, \tilde{Z}_M$. Hence $g_W(\tilde{Z}_1), \dots, g_W(\tilde{Z}_M)$ are conditionally independent given D_X . Conditioning on D_X via the law of total variance, we get

$$(6.5) \quad \text{Var}(\hat{\mathcal{R}}_{M,N}[\hat{\nu}^N]) = \mathbb{E}[\text{Var}(\hat{\mathcal{R}}_{M,N}[\hat{\nu}^N] | D_X)] + \text{Var}(\mathbb{E}[\hat{\mathcal{R}}_{M,N}[\hat{\nu}^N] | D_X]).$$

Let us observe the conditional term. Given D_X , the frequencies $\tilde{Z}_1, \dots, \tilde{Z}_M$ are i.i.d., so

$$\text{Var}(\hat{\mathcal{R}}_{M,N}[\hat{\nu}^N] | D_X) = \frac{1}{M} \text{Var}(g_W(\tilde{Z}_1) | D_X).$$

From the bound $|U_{XX}(\tilde{z})| \leq 1$, we have $|g_W(\tilde{z})| \leq \Psi(0)/2$ uniformly in $\tilde{z}$ and D_X . Hence $\text{Var}(g_W(\tilde{Z}_1) | D_X) \leq \Psi(0)^2/4$, and therefore $\mathbb{E}[\text{Var}(\hat{\mathcal{R}}_{M,N}[\hat{\nu}^N] | D_X)] = O(1/M)$.

As for the marginal term, a direct computation of the conditional expectation $\mathbb{E}[g_W(\tilde{Z}_1) | D_X]$ gives

$$\begin{aligned} \mathbb{E}[g_W(\tilde{Z}_1) | D_X] &= \frac{\Psi(0)}{2} \mathbb{E}[U_{XX}(\tilde{Z}) | D_X] \\ (6.6) \quad &= \frac{1}{2N(N-1)} \sum_{\substack{i,j=1 \\ i \neq j}}^N W(X_i, X_j) =: \bar{\mathcal{R}}_W(D_X), \end{aligned}$$

where we used $\Psi(0)\mathbb{E}[\cos(\tilde{Z}^\top(x-y))] = \Psi(x-y) = W(x,y)$ by the analogue of (2.3) for W . Thus $\mathbb{E}[\hat{\mathcal{R}}_{M,N}[\hat{\nu}^N] | D_X] = \bar{\mathcal{R}}_W(D_X)$ coincides with the kernel U -statistic estimator of $\mathcal{R}[\nu]$. It remains to bound $\text{Var}(\bar{\mathcal{R}}_W(D_X))$.

Now $\bar{\mathcal{R}}_W(D_X)$ is a one-sample U -statistic of degree 2 with symmetric kernel $\frac{1}{2}W(\cdot, \cdot)$. By Hoeffding's variance formula for one-sample U -statistics [30, Section 5.2],

$$\text{Var}(\bar{\mathcal{R}}_W(D_X)) = \frac{2}{N(N-1)} [2(N-2)\zeta_1 + \zeta_2],$$

where the Hoeffding variance components are

$$\zeta_1 = \text{Var}(\mathbb{E}[\tfrac{1}{2}W(X_1, X_2) | X_1]), \quad \zeta_2 = \text{Var}(\tfrac{1}{2}W(X_1, X_2)).$$

The uniform bound $|W(x, y)| \leq \Psi(0)$ yields $\zeta_1, \zeta_2 \leq \Psi(0)^2/4$, hence $\text{Var}(\bar{\mathcal{R}}_W(D_X)) \leq \Psi(0)^2/N$ for $N \geq 2$.

Consequently, substituting into (6.5), we obtain

$$\text{Var}(\hat{\mathcal{R}}_{M,N}[\hat{\nu}^N]) = O(1/M) + O(1/N),$$

completing the proof of the theorem. $\square$

6.3 Proof of Lemma 4.4

Decompose $\hat{\gamma}_{M_n, N_n}^2 = \bar{\gamma}_K^2(D_{N_n}) + B_{M_n}$ as in the proof of Theorem 3.1(iii) in Section 6.1, where $D_{N_n} = (X_1, \dots, X_{N_n}, Y_1, \dots, Y_{N_n})$ and

$$B_{M_n} := \frac{1}{M_n} \sum_{r=1}^{M_n} [g(Z_r) - \bar{\gamma}_K^2(D_{N_n})].$$

By Hoeffding's theorem on the strong law of large numbers for two-sample U -statistics [30, Section 5.4], $\bar{\gamma}_K^2(D_{N_n}) \rightarrow \gamma_K(\mu, \nu)^2$ as $n \rightarrow \infty$, a.s.

Since cosines and sines are bounded by 1, the Cauchy-Schwarz inequality yields $S_c(z)^2 + S_s(z)^2 \leq N^2$. Combined with (3.2)–(3.3), this gives $|U_{XX}(z)|, |V_{XY}(z)|, |U_{YY}(z)| \leq 1$ for every $z \in \mathbb{R}^d$ and every realization of the data, so

$$|g(z)| \leq \Phi(0)(|U_{XX}(z)| + 2|V_{XY}(z)| + |U_{YY}(z)|) \leq 4\Phi(0),$$

uniformly in z and D_{N_n} . In particular $|\bar{\gamma}_K^2(D_{N_n})| = |\mathbb{E}[g(Z_1) | D_{N_n}]| \leq 4\Phi(0)$.

Fix a realization of D_{N_n} and define

$$\xi_r := g(Z_r) - \bar{\gamma}_K^2(D_{N_n}), \quad r = 1, \dots, M_n.$$

Since $Z_1, \dots, Z_{M_n}$ are i.i.d. and independent of D_{N_n} , the ξ_r are, conditionally on D_{N_n} , i.i.d. with mean zero, and take values in the interval

$$J(D_{N_n}) := [-4\Phi(0) - \bar{\gamma}_K^2(D_{N_n}), 4\Phi(0) - \bar{\gamma}_K^2(D_{N_n})] \subset \mathbb{R},$$

which has the length $8\Phi(0)$. Hoeffding's inequality [15, Theorem 2] for bounded i.i.d. centered random variables therefore gives, for every $t > 0$,

$$\mathbb{P} \left(\left| \sum_{r=1}^{M_n} \xi_r \right| \geq t \mid D_{N_n} \right) \leq 2 \exp \left(-\frac{2t^2}{M_n(8\Phi(0))^2} \right).$$

Let $\varepsilon > 0$. Setting $t = M_n \varepsilon$ and dividing inside the probability by M_n , we get

$$\mathbb{P}(|B_{M_n}| \geq \varepsilon \mid D_{N_n}) \leq 2 \exp \left(-\frac{M_n \varepsilon^2}{32\Phi(0)^2} \right).$$

The right-hand side does not depend on the realization of D_{N_n} , so taking expectation over D_{N_n} preserves the bound:

$$(6.7) \quad \mathbb{P}(|B_{M_n}| \geq \varepsilon) \leq 2 \exp \left(-\frac{M_n \varepsilon^2}{32\Phi(0)^2} \right).$$

Under the rate condition $M_n/\log n \rightarrow \infty$, the series $\sum_{n \geq 1} \exp(-M_n \varepsilon^2/(32\Phi(0)^2))$ converges for every fixed $\varepsilon > 0$, because $M_n \varepsilon^2/(32\Phi(0)^2) \geq 2 \log n$ for all sufficiently large n . The first Borel–Cantelli lemma then yields $\mathbb{P}(|B_{M_n}| \geq \varepsilon \text{ infinitely often}) = 0$ for every $\varepsilon > 0$, hence $B_{M_n} \rightarrow 0$ as $n \rightarrow \infty$, a.s.

Combining the two parts, we arrive at $\hat{\gamma}_{M_n, N_n}^2 \rightarrow \gamma_K(\mu, \nu)^2$, a.s. $\square$

6.4 Proof of Theorem 4.6

The proof extends [24, Theorem 3.1], which treats the $c = 0$ specialization at the population level, in two ways: the empirical penalty problem (4.6) replaces the exact penalty problem along almost every sample path via the strong consistency established in Step (i) below, and the running interaction term $c \int_0^T \mathcal{R}[\mathcal{L}(X_t)] dt$ enters the chain of inequalities through the same path-wise concentration argument. The condition $\lambda_n \varepsilon_n \rightarrow 0$ controls the resulting approximation error.

For notational simplicity, we write

$$\mathcal{C}(u) := \mathbb{E} \left[\int_0^T |u(t, X_t)|^2 dt \right], \quad \hat{\mathcal{C}}_N(u) := \frac{1}{N} \sum_{i=1}^N \int_0^T |u(t, X_t^{(i)})|^2 dt, \quad \mathcal{R}_T(u) := \int_0^T \mathcal{R}[\mathcal{L}(X_t)] dt,$$

and we let $\hat{\mathcal{R}}_T(u)$ denote the empirical counterpart of $\mathcal{R}_T(u)$ on the time grid $\{t_k\}$ used in Algorithm 4.1, i.e., $\hat{\mathcal{R}}_T(u) := \sum_{k=1}^q \hat{\mathcal{R}}_{M,N}[\hat{\nu}_{t_k}^N](t_k - t_{k-1})$, with the number of grid points q fixed and independent of n .

Step (i): strong consistency. We show that, on a measurable set of probability one, the empirical terminal MMD², the empirical energy, and the empirical interaction integral all converge to their population counterparts, uniformly over the relevant family of laws and controls.

By Lemma 4.4, applied to each fixed pair (ν, μ_T) , and Lemma 4.5, applied to each fixed admissible control u , combined via a countable union over $n \in \mathbb{N}$, there exists a measurable set of probability one on which both $\hat{\gamma}_{M_n, N_n}^2(\nu, \mu_T) \rightarrow \gamma_K(\nu, \mu_T)^2$ and $\hat{\mathcal{C}}_{N_n}(u) \rightarrow \mathcal{C}(u)$ as $n \rightarrow \infty$, for $\nu \in \{\mu_T\} \cup \{\mathcal{L}(X_T^{(n)}) : n \geq 1\}$ and $u \in \{u^*\} \cup \{u_n : n \geq 1\}$, where u^* is an optimal control for the constrained problem $\inf\{\frac{1}{2}\mathcal{C}(u) + c\mathcal{R}_T(u) : u \in \mathcal{A}, \mathcal{L}(X_T) = \mu_T\}$.

The same Hoeffding plus Borel–Cantelli argument used in the proof of Lemma 4.4 applies to the empirical interaction $\hat{\mathcal{R}}_{M, N}[\hat{\nu}^N]$. Assumption (A) for W gives $|W(x, y)| \leq \Psi(0)$ and hence $|\hat{\mathcal{R}}_{M, N}[\hat{\nu}^N]| \leq \Psi(0)/2$ uniformly. Decomposing $\hat{\mathcal{R}}_{M_n, N_n}[\hat{\nu}^{N_n}] = \bar{\mathcal{R}}_W(D_{X, N_n}) + B_{M_n}^W$, where $\bar{\mathcal{R}}_W(D_{X, N_n}) := \frac{1}{2N_n(N_n-1)} \sum_{i \neq j} W(X_i, X_j)$ is the kernel U -statistic of W from (6.6) and $B_{M_n}^W$ is the random-feature residual, Hoeffding's inequality applied to $B_{M_n}^W$ gives $\mathbb{P}(|B_{M_n}^W| \geq \varepsilon) \leq 2\exp(-M_n\varepsilon^2/(8\Psi(0)^2))$. Combining with the SLLN for one-sample U -statistics [30, Section 5.4] applied to $\bar{\mathcal{R}}_W(D_{X, N_n})$ and the rate condition $M_n/\log n \rightarrow \infty$, we get

$$(6.8) \quad \hat{\mathcal{R}}_{M_n, N_n}[\hat{\nu}^{N_n}] \xrightarrow{\text{a.s.}} \mathcal{R}[\nu] \quad (n \rightarrow \infty),$$

for each fixed ν . A further countable union over the time grid $\{t_k\}_{k=1}^q$ and over the countable family of laws $\{\mathcal{L}(X_{t_k}^{(n)})\}_{n, k \geq 1}$ yields a measurable set Ω_0 of probability one on which all of the above limits hold simultaneously.

Let $\omega \in \Omega_0$ be fixed. By (4.8), there exists $n_0 = n_0(\omega)$ such that for $n \geq n_0$,

$$(6.9) \quad \begin{aligned} |\hat{\gamma}_{M_n, N_n}^2(\mathcal{L}(X_T^{(n)}), \mu_T) - \gamma_K(\mathcal{L}(X_T^{(n)}), \mu_T)^2| &\leq \varepsilon_n, & \hat{\gamma}_{M_n, N_n}^2(\mu_T, \mu_T) &\leq \varepsilon_n, \\ |\hat{\mathcal{C}}_{N_n}(u_n) - \mathcal{C}(u_n)| &\leq \varepsilon_n, & |\hat{\mathcal{C}}_{N_n}(u^*) - \mathcal{C}(u^*)| &\leq \varepsilon_n, \\ |\hat{\mathcal{R}}_T(u_n) - \mathcal{R}_T(u_n)| &\leq \varepsilon_n, & |\hat{\mathcal{R}}_T(u^*) - \mathcal{R}_T(u^*)| &\leq \varepsilon_n, \end{aligned}$$

where the last line uses the uniformity of the bound over the finite time grid.

Step (ii): $\sqrt{\lambda_n} \gamma_K(\mathcal{L}(X_T^{(n)}), \mu_T) \rightarrow 0$. We show that the population terminal discrepancy vanishes faster than $1/\sqrt{\lambda_n}$, almost surely. The ε_n -optimality of u_n for (4.6) gives

$$(6.10) \quad \frac{1}{2}\hat{\mathcal{C}}_{N_n}(u_n) + c\hat{\mathcal{R}}_T(u_n) + \lambda_n \hat{\gamma}_{M_n, N_n}^2(\mathcal{L}(X_T^{(n)}), \mu_T) \leq \hat{H}_{\lambda_n, c, M_n, N_n} + \varepsilon_n.$$

The optimal control u^* for the constrained MFG satisfies $\mathcal{L}(X_T^*) = \mu_T$ and $\frac{1}{2}\mathcal{C}(u^*) + c\mathcal{R}_T(u^*) = H_c^*$. Since u^* is admissible for (4.6), by (6.9),

$$(6.11) \quad \hat{H}_{\lambda_n, c, M_n, N_n} \leq \frac{1}{2}\hat{\mathcal{C}}_{N_n}(u^*) + c\hat{\mathcal{R}}_T(u^*) + \lambda_n \hat{\gamma}_{M_n, N_n}^2(\mu_T, \mu_T) \leq H_c^* + \left(\frac{1}{2} + c\right)\varepsilon_n + \lambda_n \varepsilon_n.$$

Substituting (6.11) into (6.10) and using (6.9) on the left-hand side,

$$(6.12) \quad \frac{1}{2}\mathcal{C}(u_n) + c\mathcal{R}_T(u_n) + \lambda_n \gamma_K(\mathcal{L}(X_T^{(n)}), \mu_T)^2 \leq H_c^* + \left(\frac{5}{2} + 2c\right)\varepsilon_n + 2\lambda_n \varepsilon_n.$$

Since $\lambda_n \varepsilon_n \rightarrow 0$ and $\varepsilon_n \rightarrow 0$ by (4.8), the right-hand side equals $H_c^* + o(1)$. Under (A) the kernel W is non-negative, so $\mathcal{R}[\nu] \geq 0$ for every ν and hence $\mathcal{R}_T(u_n) \geq 0$. Therefore $\lambda_n \gamma_K(\mathcal{L}(X_T^{(n)}), \mu_T)^2 \leq H_c^* + o(1)$ is bounded, and $\gamma_K(\mathcal{L}(X_T^{(n)}), \mu_T) \rightarrow 0$ since $\lambda_n \rightarrow \infty$.

It remains to upgrade this to $\lambda_n \gamma_n^2 \rightarrow 0$, where $\gamma_n := \gamma_K(\mathcal{L}(X_T^{(n)}), \mu_T)$. The key ingredient is the following tightness and lower-semicontinuity property, which is also used in Step (iii) below.

Tightness and lower-semicontinuity. By Girsanov's theorem applied to (4.1), for every $u \in \mathcal{A}$ the law $Q^u := \mathcal{L}(X^u)$ on $C([0, T]; \mathbb{R}^d)$ is absolutely continuous with respect to the reference law P^σ (under which the canonical process is a Brownian motion of variance σ^2 started from μ_0), and the relative entropy admits the identity

$$(6.13) \quad H(Q^u | P^\sigma) = \frac{1}{2\sigma^2} \mathcal{C}(u).$$

For the sequence $\{u_n\} \subset \mathcal{A}$ obtained from (4.6), the bound (6.12) together with $\mathcal{R}_T(u_n) \geq 0$ gives

$$(6.14) \quad \mathcal{C}(u_n) \leq 2H_c^* + o(1), \quad \mathcal{R}_T(u_n) \leq \frac{H_c^* + o(1)}{c} \quad (\text{when } c > 0).$$

We claim that $\{Q^{u_n}\}_{n \geq 1}$ is tight on $C([0, T]; \mathbb{R}^d)$. Decompose the SDE solution as $X_t^{(n)} = X_0 + A_t^{(n)} + \sigma B_t$ with the drift integral $A_t^{(n)} := \int_0^t u_n(r, X_r^{(n)}) dr$. The Brownian component σB has paths in $C([0, T]; \mathbb{R}^d)$, with the law of σB being Wiener measure, hence tight on $C([0, T]; \mathbb{R}^d)$. For the drift component, Cauchy–Schwarz gives $|A_t^{(n)} - A_s^{(n)}| \leq \sqrt{t-s} V_n$ with $V_n := (\int_0^T |u_n(r, X_r^{(n)})|^2 dr)^{1/2}$, so the modulus of continuity satisfies $\omega_{A^{(n)}}(\delta) \leq \sqrt{\delta} V_n$; by Markov's inequality and $\mathbb{E}V_n^2 = \mathcal{C}(u_n)$ uniformly bounded via (6.14), $\mathbb{P}(\omega_{A^{(n)}}(\delta) \geq \eta) \leq \delta \mathbb{E}V_n^2 / \eta^2 \rightarrow 0$ as $\delta \rightarrow 0$, uniformly in n . Combined with the tight initial law μ_0 and the tight Brownian component, the modulus-of-continuity criterion for tightness on $C([0, T]; \mathbb{R}^d)$ (Billingsley [4, Theorem 7.3]) yields the tightness of $\{Q^{u_n}\}$.

Let $\hat{Q}$ be any weak subsequential limit of $\{Q^{u_n}\}$, and let $\hat{Q}_t$ denote its marginal at time $t \in [0, T]$. For each fixed t , the evaluation map $\pi_t : C([0, T]; \mathbb{R}^d) \rightarrow \mathbb{R}^d$, $\omega \mapsto \omega(t)$, is continuous, so the continuous mapping theorem [4, Theorem 2.7] applied to π_t yields $\mathcal{L}(X_t^{(n)}) = Q^{u_n} \circ \pi_t^{-1} \rightarrow \hat{Q} \circ \pi_t^{-1} = \hat{Q}_t$ weakly on $\mathcal{P}(\mathbb{R}^d)$ along the subsequence. At $t = T$, combined with $\gamma_n \rightarrow 0$ and the fact that γ_K metrizes the weak topology on $\mathcal{P}(\mathbb{R}^d)$ under (A) (Remark 2.1),

$$(6.15) \quad \hat{Q}_T = \mu_T.$$

By the lower-semicontinuity of relative entropy under the weak topology (see, e.g., [13, Lemma 1.4.3]) and (6.13),

$$(6.16) \quad \frac{1}{2} \mathcal{C}(\hat{u}) := \sigma^2 H(\hat{Q} | P^\sigma) \leq \sigma^2 \liminf_{n \rightarrow \infty} H(Q^{u_n} | P^\sigma) = \frac{1}{2} \liminf_{n \rightarrow \infty} \mathcal{C}(u_n),$$

where $\hat{u}$ denotes the drift corresponding to $\hat{Q}$. Moreover, since $W \in C_b(\mathbb{R}^d \times \mathbb{R}^d)$ under (A), the functional $\nu \mapsto \mathcal{R}[\nu] = \frac{1}{2} \iint W(x, y) \nu(dx) \nu(dy)$ is continuous on $\mathcal{P}(\mathbb{R}^d)$, and the marginal convergence $\mathcal{L}(X_t^{(n)}) \rightarrow \hat{Q}_t$ established above yields $\mathcal{R}[\mathcal{L}(X_t^{(n)})] \rightarrow \mathcal{R}[\hat{Q}_t]$ for every $t \in [0, T]$. Combined with the uniform bound $|\mathcal{R}[\mathcal{L}(X_t^{(n)})]| \leq \Psi(0)/2$ and bounded convergence on $[0, T]$,

$$(6.17) \quad \mathcal{R}_T(u_n) \rightarrow \mathcal{R}_T(\hat{u}) \quad (n \rightarrow \infty, \text{ along the subsequence}).$$

Combining (6.15), (6.16), (6.17), and the constrained definition of H_c^* ,

$$(6.18) \quad H_c^* \leq \frac{1}{2}\mathcal{C}(\hat{u}) + c\mathcal{R}_T(\hat{u}) \leq \liminf_{n \rightarrow \infty} \left[\frac{1}{2}\mathcal{C}(u_n) + c\mathcal{R}_T(u_n) \right]$$

along every weakly convergent subsequence. A standard sub-subsequence argument extends (6.18) to the full sequence.

Conclusion of Step (ii). Suppose, for contradiction, that $\limsup_n \lambda_n \gamma_n^2 \geq 5\delta$ for some $\delta > 0$. Extract a subsequence (still denoted $\{n\}$) along which $\lambda_n \gamma_n^2 \geq 4\delta$ for all sufficiently large n . From (6.12),

$$\frac{1}{2}\mathcal{C}(u_n) + c\mathcal{R}_T(u_n) \leq H_c^* - 4\delta + o(1) \quad \text{along the subsequence.}$$

Applying (6.18) along the same subsequence,

$$H_c^* \leq \liminf_{n \rightarrow \infty} \left[\frac{1}{2}\mathcal{C}(u_n) + c\mathcal{R}_T(u_n) \right] \leq H_c^* - 4\delta,$$

which is impossible. Hence $\lim_{n \rightarrow \infty} \lambda_n \gamma_n^2 = 0$, i.e., $\sqrt{\lambda_n} \gamma_K(\mathcal{L}(X_T^{(n)}), \mu_T) \rightarrow 0$, which proves conclusion (i) of Theorem 4.6.

Step (iii): $\hat{H}_{\lambda_n, c, M_n, N_n} \rightarrow H_c^*$. We show that the empirical penalty value converges to the constrained MFG value, almost surely.

The upper bound $\limsup_{n \rightarrow \infty} \hat{H}_{\lambda_n, c, M_n, N_n} \leq H_c^*$ follows from (6.11) together with $\lambda_n \varepsilon_n \rightarrow 0$ and $\varepsilon_n \rightarrow 0$.

For the lower bound, (6.10) together with $\hat{\gamma}_{M_n, N_n}^2 \geq 0$ gives

$$\hat{H}_{\lambda_n, c, M_n, N_n} + \varepsilon_n \geq \frac{1}{2}\hat{\mathcal{C}}_{N_n}(u_n) + c\hat{\mathcal{R}}_T(u_n).$$

By (6.9), $\hat{\mathcal{C}}_{N_n}(u_n) \geq \mathcal{C}(u_n) - \varepsilon_n$ and $\hat{\mathcal{R}}_T(u_n) \geq \mathcal{R}_T(u_n) - \varepsilon_n$, so

$$\hat{H}_{\lambda_n, c, M_n, N_n} + (\frac{3}{2} + c)\varepsilon_n \geq \frac{1}{2}\mathcal{C}(u_n) + c\mathcal{R}_T(u_n).$$

Taking $\liminf_n$ and applying (6.18),

$$\liminf_{n \rightarrow \infty} \hat{H}_{\lambda_n, c, M_n, N_n} \geq \liminf_{n \rightarrow \infty} \left[\frac{1}{2}\mathcal{C}(u_n) + c\mathcal{R}_T(u_n) \right] \geq H_c^*.$$

Combined with the upper bound, $\lim_n \hat{H}_{\lambda_n, c, M_n, N_n} = H_c^*$, which proves conclusion (ii) of Theorem 4.6. This completes the proof. $\square$

Remark 6.2. By Lemma 4.4, the strong law gives $\hat{\gamma}_{M, N}^2 - \gamma_K^2 = o(1)$ a.s. as $M, N \rightarrow \infty$, so for any deterministic sequences $M_n, N_n \rightarrow \infty$ the path-wise rate of this convergence may be absorbed into ε_n without imposing additional quantitative conditions on (M_n, N_n) . This contrasts with approaches based on biased random-feature estimators (cf. Remark 3.2), which require explicit control of the approximation error and an extra penalty correction. In the stochastic optimization setting of Algorithm 4.1, quantitative bounds on ε_n follow from the variance estimate of Theorem 3.1(iii): the in-probability counterpart of $\lambda_n \varepsilon_n \rightarrow 0$ amounts to $\lambda_n / \sqrt{M_n} \rightarrow 0$ and $\lambda_n / \sqrt{N_n} \rightarrow 0$.

6.5 Proof of Proposition 4.8

The proof of Theorem 4.6 given in Section 6.4 uses three structural properties of the running cost: non-negativity, weak continuity, and a.s. consistency of the empirical estimator. All three are now assumed directly: the consistency hypothesis (4.9) replaces the consistency obtained in Step (i) via Hoeffding's inequality and Borel–Cantelli applied to the random-feature residual $B_{M_n}^W$, and Steps (ii)–(iii) carry over verbatim with $\hat{\mathcal{R}}_{M_n, N_n}$ replaced by $\hat{\mathcal{R}}_{N_n}$. $\square$

6.6 Proof of Proposition 4.9

We refine the path-wise inequalities (6.9) in the proof of Theorem 4.6 given in Section 6.4 to have explicit Hoeffding rates, and substitute the refined bounds into the chain (6.11)–(6.12).

Step (i): rate for $\hat{\gamma}_{M_n, N_n}^2$. We establish a quantitative Hoeffding rate for the random-feature MMD² estimator, which refines the qualitative input (6.9).

By the Hoeffding bound (6.7) in the proof of Lemma 4.4, applied to the random-feature residual B_{M_n} conditional on the data,

$$\mathbb{P}(|B_{M_n}| \geq t) \leq 2 \exp\left(-\frac{M_n t^2}{32\Phi(0)^2}\right), \quad t > 0.$$

Setting $t = 8\Phi(0)\sqrt{2\log n/M_n}$ gives $\mathbb{P}(|B_{M_n}| \geq t) \leq 2n^{-2}$, which is summable. So the Borel–Cantelli lemma yields

$$(6.19) \quad |B_{M_n}| \leq 8\Phi(0)\sqrt{2\log n/M_n} \quad \text{for all sufficiently large } n, \text{ a.s.}$$

Next, $\bar{\gamma}_K^2(D_{N_n})$ is the two-sample U -statistic with kernel $h(x, x', y, y') := K(x, x') - K(x, y') - K(y, x') + K(y, y')$ bounded by $4\Phi(0)$. By Hoeffding's inequality for two-sample U -statistics [30, Theorem 5.6.1.A], for any fixed pair (μ, ν) and any $t > 0$,

$$\mathbb{P}(|\bar{\gamma}_K^2(D_{N_n}) - \gamma_K(\mu, \nu)^2| \geq t) \leq 2 \exp\left(-\frac{N_n t^2}{32\Phi(0)^2}\right).$$

The same Borel–Cantelli argument gives, for every fixed pair (μ, ν) ,

$$(6.20) \quad |\bar{\gamma}_K^2(D_{N_n}) - \gamma_K(\mu, \nu)^2| \leq 8\Phi(0)\sqrt{2\log n/N_n} \quad \text{for all sufficiently large } n, \text{ a.s.}$$

Under (H2), the samples at iteration n are independent of u_n , so we may apply (6.19)–(6.20) conditionally on u_n to the pair $(\nu, \mu_1) = (\mathcal{L}(X_1^{(n)}), \mu_1)$ and to the pair (μ_1, μ_1) , for which $\gamma_K^2 = 0$. Combining via $\hat{\gamma}_{M_n, N_n}^2 = \bar{\gamma}_K^2(D_{N_n}) + B_{M_n}$ and taking a countable union over $n \geq 1$, we obtain a measurable event Ω_1 of probability one such that for $\omega \in \Omega_1$,

$$(6.21) \quad \begin{aligned} |\hat{\gamma}_{M_n, N_n}^2(\mathcal{L}(X_1^{(n)}), \mu_1) - \gamma_K(\mathcal{L}(X_1^{(n)}), \mu_1)^2| &\leq \delta_n^\gamma, \\ \hat{\gamma}_{M_n, N_n}^2(\mu_1, \mu_1) &\leq \delta_n^\gamma, \end{aligned}$$

for all sufficiently large n , where $\delta_n^\gamma := 16\Phi(0)(\sqrt{2\log n/M_n} + \sqrt{2\log n/N_n})$.

Step (ii): rate for $\hat{\mathcal{C}}_{N_n}$. We establish a Hoeffding rate for the empirical energy.

Recall the population energy and its empirical counterpart on N samples, $\mathcal{C}(u) := \mathbb{E}[\int_0^1 |u(t, X_t)|^2 dt]$ and $\hat{\mathcal{C}}_N(u) := \frac{1}{N} \sum_{i=1}^N \int_0^1 |u(t, X_t^{(i)})|^2 dt$, introduced in Section 6.4 with $T = 1$. Under (H1), the integrand $\int_0^1 |u(t, X_t)|^2 dt$ is bounded by L^2 for $u \in \{u^*, u_n\}$ and for every path. Hoeffding's inequality for sums of bounded i.i.d. random variables gives, for every $t > 0$,

$$\mathbb{P}(|\hat{\mathcal{C}}_{N_n}(u) - \mathcal{C}(u)| \geq t) \leq 2 \exp\left(-\frac{2N_n t^2}{L^4}\right).$$

By (H2), u_n is independent of the samples at iteration n , so the same Borel–Cantelli argument applies conditionally on u_n and yields

$$(6.22) \quad |\hat{\mathcal{C}}_{N_n}(u_n) - \mathcal{C}(u_n)| \vee |\hat{\mathcal{C}}_{N_n}(u^*) - \mathcal{C}(u^*)| \leq \delta_n^{\mathcal{C}} \quad \text{for all sufficiently large } n, \text{ a.s.,}$$

where $\delta_n^{\mathcal{C}} := L^2 \sqrt{2 \log n / N_n}$.

Step (iii): assembly. We combine the bounds of Steps (i)–(ii) with the chain (6.11)–(6.12) to obtain (4.10) and (4.11).

Working on Ω_1 intersected with the analogous event from Step (ii), both events of probability one, we substitute (6.21) and (6.22) for the qualitative input (6.9) in the chain (6.11)–(6.12) of the proof of Theorem 4.6, keeping the empirical quantities on both sides. The upper bound (6.11) becomes $\hat{H}_{\lambda_n, M_n, N_n} \leq H^* + \frac{1}{2} \delta_n^{\mathcal{C}} + \lambda_n \delta_n^{\gamma}$, and combining with the ε_n -optimality (6.10) of u_n yields, after dropping the nonnegative term $\frac{1}{2} \hat{\mathcal{C}}_{N_n}(u_n) \geq 0$ on the left-hand side,

$$(6.23) \quad \lambda_n \hat{\gamma}_{M_n, N_n}^2(\mathcal{L}(X_1^{(n)}), \mu_1) \leq H^* + \varepsilon_n + \frac{1}{2} \delta_n^{\mathcal{C}} + \lambda_n \delta_n^{\gamma}.$$

Since both $\delta_n^{\mathcal{C}}$ and δ_n^{γ} carry the factor $\sqrt{\log n / N_n}$ and $\lambda_n \rightarrow \infty$, the ratio $\delta_n^{\mathcal{C}} / \lambda_n$ is dominated by δ_n^{γ} . Dividing (6.23) by λ_n then yields (4.10) with a constant C^* that may be taken as $C^* = 17\Phi(0) + L^2$. The population-level bound (4.11) now follows by adding δ_n^{γ} to the left-hand side of (6.23) via $\hat{\gamma}_{M_n, N_n}^2(\mathcal{L}(X_1^{(n)}), \mu_1) \geq \gamma_K(\mathcal{L}(X_1^{(n)}), \mu_1)^2 - \delta_n^{\gamma}$ from (6.21), which absorbs into the same $\sqrt{\log n / M_n} + \sqrt{\log n / N_n}$ term at the cost of an extra factor 2 on the leading constant. The asymptotic rate under the polynomial schedule follows by direct substitution. $\square$

6.7 Proof of Proposition 4.11

The argument of Proposition 4.9 extends to the MFG case once one adds a Hoeffding rate for the empirical interaction $\hat{\mathcal{R}}_{M_n, N_n}[\hat{\nu}_t^{N_n}]$ that parallels (6.19)–(6.21). Decompose $\hat{\mathcal{R}}_{M_n, N_n}[\hat{\nu}_t^{N_n}] = \bar{\mathcal{R}}_W(D_{X_t, N_n}) + B_{M_n, t}^W$ as in (6.6), where $|\bar{\mathcal{R}}_W| \leq \Psi(0)/2$ and $|B_{M_n, t}^W| \leq \Psi(0)$. Hoeffding's inequality applied to $B_{M_n, t}^W$ conditionally on the data gives $\mathbb{P}(|B_{M_n, t}^W| \geq \varepsilon) \leq 2 \exp(-M_n \varepsilon^2 / (2\Psi(0)^2))$, and Hoeffding's inequality for the one-sample U -statistic $\bar{\mathcal{R}}_W$ gives $\mathbb{P}(|\bar{\mathcal{R}}_W(D_{X_t, N_n}) - \mathcal{R}[\mathcal{L}(X_t^{(n)})]| \geq \varepsilon) \leq 2 \exp(-N_n \varepsilon^2 / (2\Psi(0)^2))$ (see, e.g., [30, Section 5.6]). Setting $\varepsilon = 2\Psi(0)\sqrt{2 \log n / M_n}$ and $\varepsilon = 2\Psi(0)\sqrt{2 \log n / N_n}$, respectively, and applying Borel–Cantelli as in Step (i) of Section 6.6,

$$(6.24) \quad |\hat{\mathcal{R}}_{M_n, N_n}[\hat{\nu}_t^{N_n}] - \mathcal{R}[\mathcal{L}(X_t^{(n)})]| \leq \delta_n^{\mathcal{R}} := 4\Psi(0) \left(\sqrt{\frac{2 \log n}{M_n}} + \sqrt{\frac{2 \log n}{N_n}} \right)$$

for all sufficiently large n and every fixed $t \in [0, T]$, almost surely. Taking a countable union over the finite time grid $\{t_k\}_{k=1}^q$ used in Algorithm 4.1 preserves the rate, and time-integration over $[0, T]$ multiplies $\delta_n^{\mathcal{R}}$ by T :

$$|\hat{\mathcal{R}}_T(u) - \mathcal{R}_T(u)| \leq T\delta_n^{\mathcal{R}}$$

for $u \in \{u^*, u_n\}$, almost surely for all sufficiently large n .

Substituting this bound and (6.21), (6.22) into the chain (6.11)–(6.12) of the proof of Theorem 4.6 yields, after dropping the non-negative terms,

$$\lambda_n \hat{\gamma}_{M_n, N_n}^2(\mathcal{L}(X_T^{(n)}), \mu_T) \leq H_c^* + \varepsilon_n + \frac{1}{2}\delta_n^{\mathcal{C}} + cT\delta_n^{\mathcal{R}} + \lambda_n\delta_n^{\gamma}.$$

The same argument as in Step (iii) of Section 6.6 (dividing by λ_n and absorbing $\delta_n^{\mathcal{C}}/\lambda_n$ and $cT\delta_n^{\mathcal{R}}/\lambda_n$ into δ_n^{γ}) produces (4.12) with $C_c^* = 17\Phi(0) + L^2 + 8cT\Psi(0)$. The polynomial-schedule rate follows by direct substitution. $\square$

6.8 Proof of Corollary 4.12

We follow the proof of Theorem 4.6 given in Section 6.4 with $\hat{\gamma}_{M_n, N_n}^2$ replaced by $\bar{\gamma}_K^2(D_{N_n})$ throughout and $\hat{H}_{\lambda_n, M_n, N_n}$ replaced by $\hat{H}_{\lambda_n, N_n}^{\ker}$. Only the input from Lemma 4.4 needs to be modified, and the energy estimate Lemma 4.5 is used unchanged.

The decomposition (6.3) shows that $\bar{\gamma}_K^2(D_{N_n}) = \mathbb{E}[\hat{\gamma}_{M_n, N_n}^2 | D_{N_n}]$ for every $M_n \geq 1$. In particular, $\bar{\gamma}_K^2(D_{N_n})$ is a deterministic functional of the data alone and does not involve the random Fourier features. By the strong law of large numbers for two-sample U -statistics [30, Section 5.4], we have

$$\bar{\gamma}_K^2(D_{N_n}) \longrightarrow \gamma_K(\mu, \nu)^2, \quad n \rightarrow \infty, \quad \text{a.s.},$$

under the single condition $N_n \rightarrow \infty$. This is exactly the first half of the proof of Lemma 4.4, and the random-feature residual B_{M_n} does not appear here. Consequently, the Hoeffding–Borel–Cantelli argument used in the proof of Lemma 4.4 to control B_{M_n} , which is the source of the rate condition $M_n/\log n \rightarrow \infty$ in Theorem 4.6, is no longer needed, and no condition on a random-feature count is imposed in the statement.

With this substitution, the path-wise inequalities corresponding to (6.9) in the proof of Theorem 4.6 become the following: on a measurable set of probability one, for every sufficiently large n and for the relevant pairs (ν, μ_1) used in the proof,

$$|\bar{\gamma}_K^2(D_{N_n}) - \gamma_K(\nu, \mu_1)^2| \leq \varepsilon_n, \quad \bar{\gamma}_K^2(D_{N_n})|_{\nu=\mu_1} \leq \varepsilon_n,$$

together with the energy bounds from Lemma 4.5. The remaining steps of the proof of Theorem 4.6, i.e., the upper bound (6.11), the contradiction argument in Step (ii), and the tightness-based liminf argument in Step (iii), use only these path-wise inequalities and the ε_n -optimality of u_n , and apply verbatim to the kernel U -statistic penalty problem (4.14). $\square$

7 Conclusion

We have proposed a computational framework for the soft-target version of *mean-field optimal transport*, namely a potential / planning MFG where both the running interaction cost and the terminal

distributional cost are expressed through reproducing-kernel maximum mean discrepancies, and applied it to a coordinated electric-vehicle charging problem. The framework is sample-linear (per-iteration cost $O(NM)$ in the batch size N and the random-feature count M) and dimension-agnostic within the per-iteration cost, which makes it compatible with the high-dimensional Schrödinger-bridge experiments at $d \leq 100$ of Section 5.3 while remaining moderate in the dimension of the MFG application reported here. By combining the Fourier representation of the squared MMD and of the kernel self-interaction with U -statistic decompositions, we obtained unbiased estimators $\hat{\gamma}_{M,N}^2$ (Theorem 3.1) and $\hat{\mathcal{R}}_{M,N}[\hat{\nu}^N]$ (Theorem 3.3) of complexity $O(NM)$ for the terminal MMD penalty and the interaction cost respectively. Both estimators have an explicit variance decomposition $O(1/M) + O(1/N)$, which directly underlies the convergence analysis of Section 4.2.

The theoretical content of the paper is summarized by four results:

- (1) Two unbiased $O(NM)$ estimators of the kernel costs of (4.3), each with explicit variance decomposition $O(1/M) + O(1/N)$ (Theorems 3.1 and 3.3).
- (2) A sample-level almost-sure convergence theorem for the empirical potential MFG penalty problem, coupling $\lambda_n, M_n, N_n, \varepsilon_n$ through explicit rate conditions including the random-feature condition $M_n/\log n \rightarrow \infty$ (Theorem 4.6), with the Schrödinger-bridge specialization recovered as Corollary 4.7.
- (3) An explicit almost-sure rate of convergence of the form $\leq (H^* + \varepsilon_n)/\lambda_n + O(\sqrt{\log n/M_n} + \sqrt{\log n/N_n})$ holding for all sufficiently large n , for both the observable empirical penalty and the population γ_K^2 , yielding an algebraic rate $\gamma_K = O(n^{-a/2}(\log n)^{1/4})$ a.s. under polynomial schedules $\lambda_n = n^a, M_n = N_n = n^{2a}$ (Proposition 4.9).
- (4) Recovery of the Schrödinger-bridge problem of [24] as the special case $\mathcal{R} \equiv 0$ (Corollary 4.7), together with the first sample-level convergence guarantee for the empirical kernel- U -statistic implementation of that method (Corollary 4.12).

Numerical experiments demonstrated the framework on two fronts: the Schrödinger-bridge specialization ($\mathcal{R} \equiv 0$) in dimensions up to $d = 100$ with Gaussian shift and bimodal targets (Sections 5.2–5.3) and the supporting convergence diagnostics and computational-scaling experiments (Sections 5.4, 5.5); and the EV charging fleet potential MFG with per-vehicle physical heterogeneity ($c > 0$) in Section 5.6, where the aggregate-demand congestion cost $\mathcal{R}^{(D)}[\nu] = (\int \eta(h)u^*(s)d\nu)^2$ reduces peak and time-averaged aggregate demand while keeping the deadline SOC mean close to the target; the convergence guarantee in this non-kernel-cost setting follows from Proposition 4.8.

Unbiasedness in practice and in theory. The biased RFF V -statistic and the unbiased RF U -statistic terminal penalty produce empirically indistinguishable trained drifts across the configurations of Section 5.1.4. We nevertheless adopt the unbiased construction for theoretical reasons: it eliminates the $O(\Phi(0)/N)$ population-level penalty floor and yields the clean variance decomposition $O(1/M) + O(1/N)$ on which Proposition 4.9 relies.

Future directions. Several directions are worth pursuing. First, the EV demonstration uses a single physical heterogeneity coordinate; enriching h with additional vehicle-level parameters (battery capacity, plug-out time uncertainty, comfort preference) under h -dependent dynamics and a

domain-justified distribution is immediate, since the $O(N)$ and $O(NM)$ scalings of our estimators carry over. Second, richer price-feedback models (piecewise-linear price impact, grid capacity constraints) fit within Proposition 4.8 through additional terms in $\mathcal{R}[\nu_t]$ or constraints on the admissible drift class. Third, the adaptive bandwidth scaling $\alpha = 1/d$ could be replaced by deep kernel learning where the kernel operates in a learned feature space of lower effective dimension, which would be useful for scaling to higher heterogeneity dimensions. Fourth, other smart-grid problems (distributed battery storage, demand response, day-ahead market coordination) and non-energy MFGs (crowd dynamics, opinion formation, multi-agent reinforcement learning) share the structural form of (1.1)–(1.2) and admit the same treatment. Fifth, extending Proposition 4.8 to weakly lower-semicontinuous running costs would accommodate local-density penalties $\int F(\rho) dx$ arising in HJB–Fokker–Planck formulations; this requires either a stronger consistency mode or an upgrade of the lsc step in the proof of Theorem 4.6. Sixth, a head-to-head numerical benchmark against deep-learning MFG/SBP baselines on a common problem instance would complement the structural comparison of Section 5.7.

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